

Effects of Cognitive Instructions for Discerning Analytical and Intuitive Tasks

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Abstract

Human behavior has shown many examples where people made poor decisions impacting themselves and others and wasting valuable resources and time. People are prone to error when their biases influence their decision making. They make errors when analytical biases result in over thinking or their intuitive biases result in impulsive unfounded decisions. Prior research examined the use of training and instruction in determining when decision tasks require analytical thinking. However, researchers had not considered intuitive instructions as many assumed intuition to be automatic. This study investigated how instructions for discerning both analytical and intuitive decision tasks affected participants' choice of thinking style for problem resolution. The research method of this study was a quantitative, post-test only with control group, experimental design, which used a treatment group. This experimenter randomly selected 102 adult managers within the United States. This experimenter randomly assigned participants to either an intervention group or a control group. Both groups received a standard set of instructions for completing a survey. Only the intervention group received instructions for discerning between analytical and intuitive decision tasks as the intervention. Participants then completed a series of 10 word problems testing their choice of system thinking style. After each word problem, participants indicated which style of thinking they chose to use in resolving the word problem. This researcher used a t-test to analyze participants' responses to understand how instructions for discerning analytical and intuitive decision tasks affected participants' choice of thinking style for problem resolution.

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Chapter 1: Introduction

Human behaviors have shown many examples where people made poor decisions impacting themselves and others which wasted valuable resources and time (Bukhszar, 2003; Cross, Thomas, & Light, 2009; McEnroe, 2010). Many of these poor decisions have led to scandals such as Enron, WorldCom, and others leading the United States government to enforce the Sarbanes-Oxley Act (McEnroe, 2010). In the UK, poor police discretion and relying upon intuition has led to bias and prejudice in traffic stops and dangerous car chases which put the public at risk (Allen, 2011). Others have made poor decisions which have threatened their health through alcoholism and substance abuse, resulting in stays in recovery centers and sometimes prison (Gibbons, Houlihan, & Gerrard, 2009; Goldsmith & Cyboran, 2013; A. C. Moss & Albery, 2009; Yechiam et al., 2008). Health, financial success, and well-being depend upon the effectiveness and efficiency of analytical decision making (Boyle et al., 2012; Gibbons et al., 2009; Wiers, Havermans, Deutsch, & Stacy, 2008).

Dual System Theory (DST) (Kahneman, 2011) explains two different systems of human thinking in decision making as either intuitive or analytical. Understanding these two systems is the beginning of successful decision making (Evans, 2003; Kahneman, 2011). The intuitive system one thinking is effortless, fast, impulsive, and develops from conditioning through past feelings, experiences, and knowledge. The analytical system two thinking is effortful, slow, logical, and develops from training in analysis of information and logic. People experience problems with system one thinking when they are impulsive, reactionary, passion-centered, subjective, and form belief biases. People experience problems with system two thinking when they are unfriendly, impersonal,

take too much time to decide, and lack emotional considerations. Advantages of system one thinking are quick results, friendliness, and creativity (Böckenholt, 2012; Cools & Van Den Broeck, 2008; Glöckner & Witteman, 2010; Klein & Weiss, 2007). Advantages of system two thinking are reasonableness, accuracy, and objectivity (Andrews, 2010; De Neys & Glumicic, 2008; Frederick, 2005; Kogler & Kühberger, 2007; Stupple, Ball, & Ellis, 2013; Stupple, Ball, Evans, & Kamal-Smith, 2011).

Background

Both systems of thinking have appropriate purposes and value when applied to the correct decision task type (Klein & Weiss, 2007). People may use either system of thinking to correct the other during the decision process (Andrews, 2010; De Neys & Glumicic, 2008; Stupple et al., 2011). System one is most appropriate for decision tasks which require personal preferences, quick reaction, and prior conditioning (Cools & Van Den Broeck, 2008). System two is most appropriate for tasks which require careful consideration, data gathering, reasoning, and analysis (Kogler & Kühberger, 2007).

Research from Kogler and Kühberger (2007) supported that when people used their intuitive thinking for analytical decision tasks, their accuracy decreased due to their biases. Results from Cools and Van Den Broeck (2008) showed that those who used analytical thinking for intuitive decision decreased accuracy due to a prolonged or overthought responses. Kogler and Kühberger discovered when people followed instructions for identifying analytical decision tasks, they improved their decision accuracy. However, the researchers did not test instructions for identifying intuitive decisions tasks. This study extended prior research by testing how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking

style for their problem resolution.

Statement of the Problem

Few studies addressed how training and instruction improved choice of thinking style (Barton, Fugelsang, & Smilek, 2009; De Neys & Glumicic, 2008; Stuppel et al., 2013). Without instructions people tend to establish their beliefs first (Andrews, 2010; Brisson, de Chantal, Forgues, & Markovits, 2014; Thompson & Johnson, 2014). Without training, when people started to use system two reasoning, they abandoned it for their biases as task complexity increased or time limitations decreased (Brisson et al., 2014; Stuppel et al., 2013). People who inappropriately forced analytical thinking for intuitive decisions stifled creativity (Norris & Epstein, 2011).

When researchers (De Neys & Glumicic, 2008) instructed students to read problems aloud they expected it would improve their ability to answer analytical problems correctly. They anticipated that these instructions would help them to recognize they needed an analytical thinking style to respond, which would improve their response accuracy. Instead, reading aloud had no effect on the participants' ability to identify the decision task as an analytical problem. When Kogler and Kühberger (2007) experimented with instructions for identifying analytical type decision tasks, they discovered that people improved their accuracy when they recognized the decision task required analytical thinking.

The problem was in understanding how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for problem resolution. People were prone to error when their biases influenced their thinking style for decision tasks (Andrews, 2010; Goldsmith & Cyboran, 2013; Kahneman, 2011;

Kogler & Kühberger, 2007; Stupple et al., 2011). Prior, researchers had only investigated analytical type decision tasks and did not provide any investigation of instructions for intuitive decisions (Barton et al., 2009; De Neys & Glumicic, 2008; Kogler & Kühberger, 2007). While they recognized the importance of people identifying and using the correct system thinking style in reducing decision errors, they assumed only analytical instructions were necessary. Their research supported their hypothesis that instructions for identifying analytical type decision tasks improved participants' decision accuracy. These studies measured correct answers, but not participant's choice to think analytically. Decision accuracy was problematic in that participants must have had skills to resolve tasks correctly (Thompson & Johnson, 2014). Some chose the correct style of thinking but still answered the question incorrectly.

Purpose of the Study

The purpose of this quantitative study was to gain understanding in how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for their problem resolution. This quantitative experimental study used a post-test only with control group design as defined by de Vaus (2001) and specified by Black (1999). Because this researcher's area was associated with industrial and organizational psychology, the influence of managerial decision making was the primary focus as other researchers have shown (Cerni, Curtis, & Colmar, 2012; Curtis & Lee, 2013; Ralf, Spang, & Ozcan, 2009; Seiler, Fischer, & Ooi, 2010). Qualtrics (2014) had the feature of allowing researchers to identify a target audience from Qualtrics's panel of which it administered as an anonymous survey for this researcher. This researcher used Qualtrics's targeting feature to randomly select participants. From this

list, at least 102 adult managers from across the United States in any organization participated with an effect size of 0.5. For this study, participants were of any age beyond 18 years as long as they are actively engaged in a management role. Management adults were those who were responsible for people who directly reported to them and were the decision makers for an organization. This researcher used Qualtrics to randomly assign participants to either an intervention group or a control group which ensured probabilistic equivalence. The survey provided Intervention participants with special instructions as an intervention on how to discern between analytical and intuitive types of decisions, as used by Kogler and Kühberger (2007). The survey only provided the control group with the standard instructions for completing a survey. Both groups received a standard set of instructions on how to complete a survey. The independent variable was the group type which was randomly assigned to participants. Both groups completed the same decision making tasks taken from De Neys and Glumicic (2008). As the dependent variable, after each decision task, participants were asked to identify if they used either intuitive thinking style (“I relied on my intuitive impressions,”) or analytical thinking style (“I had to think things through carefully”) which was based upon the Rational Experiential Inventory (REI) (Pacini & Epstein, 1999).

Theoretical Framework

People think in two different ways (Kahneman, 2011; Witteman, van den Bercken, Claes, & Godoy, 2009). Sometimes they think quickly and sometimes they think slowly. Dual system theory (DST) is a meso-theory which explains these two different ways of thinking. The quick way in which people think takes little effort for it is intuitive and automatic. The slow way in which people think requires much effort with

reasoning and applying rules. Dual process theory (DPT) is very similar to DST and researchers interchange these theories because of similarity (Evans, 2012).

How theory explains human behavior.

People are born with intuitive thinking ability and this ability continues to develop throughout their lives (Spezio, 2011). People develop their system one intuitive thinking through both conscious and unconscious means as they experience life through their impressions and feelings. They retain their intuitive thinking even after system two analytical thinking is gone (Boyle et al., 2012). People use their system one thinking to make snap decisions that are often based upon their belief-bias, feelings, or passion (Andrews, 2010; Stupple et al., 2013).

People have to make a conscious and deliberate effort to develop and use system two analytical thinking style (Kahneman, 2011). The prefrontal cortex (PFC) is the location of the brain that controls analytical thinking (Spezio, 2011). From this part of the brain, people are able to reason, monitor their behavior, and take corrective actions in decision making. De Neys and Glumicic (2008) claimed that 80% of educated adults make poor decisions when faced with conflicting information. They used DPT to help understand how people make these thinking failures. They contend that people make the wrong decision because of their bias even when presented with evidence to the contrary.

Controversy and Unanswered Questions

The difference between DST and DPT is that Kahneman (2011) described the way people think as two systems while Evans (2012) referred to two types of thinking processes. Evans' issue with system thinking was that people misunderstood the terminology and thought that there are only two specific systems of thinking. Kahneman

preferred two systems over processes as they referred to different sections of the brain that produced intuitive and deliberate analytical thinking. Kahneman saw the thinking and decision process working through these two systems. Evans held to systems one and two being made up of many diverse processes that he categorized into two types of intuitive and reflective while Kahneman referred to them as intuitive and deliberate. Both agreed that understanding these two systems or processing types was important for successful decision making.

De Neys and Glumicic (2008) explained that there was still a dispute on whether system two monitors system one and overrides the intuitive decision with reasoning or system one overrides system two with internal feelings. Most people never realize that one system monitors the other, overrides the other, or causes a change of mind. People find these situations even more difficult when confronted with evidence that is contrary to their personal belief-biases (Andrews, 2010). Biases are an inclination to use emotions, past experiences, and past impressions in decision making. These biases override logic and evidence. System two also has biases, but they are more complex. These biases use system two thinking to procrastinate, overvalue benefits, undervalue consequences, or selectively include only favorable evidence (Goldsmith & Cyboran, 2013).

Because of these conflicts, De Neys and Glumicic (2008) experimented with influencing subjects' choice through system type identification. In their study, researchers instructed participants to read aloud decision questions in order to recognize the system two type decision tasks and respond accordingly with analytical thinking. They expected their participants would see and hear that system two analytical thinking was needed for the analytical problems. This would cause the participants to realize that they must think

through the decision task analytically and answer the task correctly but their results did not support their prediction. Reading the questions out loud did not improve the participants' decision accuracy.

Kogler and Kühberger (2007) also experimented with system type identification by providing their participants with system two identifying instructions. Their expectation was that if they informed the participants to look for clues, such as statistics, they would recognize that they needed to apply a system two analytical thinking style and make the correct decision. Their findings were more promising but neither of these two studies provided any system one identification instructions for an intervention.

Both of these studies (De Neys & Glumicic, 2008; Kogler & Kühberger, 2007) focused on system two only, assuming that system one intuitive thinking would be automatic for intuitive decision tasks. Prior research consistently neglected system one intuitive instructions. Additionally, these studies were only focused on whether participants answered decision tasks correctly, not if the appropriate system thinking style was selected. The theoretical gap in DST and DPT was understanding how instructions, which determined differences between analytical and intuitive decision tasks, influenced a person's choice of thinking style in resolving decision tasks. When people realized that a decision task requires analytical thinking style, they were more likely to choose and use analytical thinking in responding to the decision task and thus improved their decision accuracy (Andrews, 2010; Goldsmith & Cyboran, 2013; Kahneman, 2011; Kogler & Kühberger, 2007; Stupple et al., 2011).

Discussion

Within DST and DPT there are theories that relate to system one belief biases

(Andrews, 2010) and system two analytical biases (Barton et al., 2009; Goldsmith & Cyboran, 2013). These biases create a dilemma for people when they override the appropriate choice and use of system thinking for a decision task. Researchers have had some success with influencing these biases through identification of system two decision types (De Neys & Glumicic, 2008; Kogler & Kühberger, 2007). Through this research, results supported that identifying analytical decision tasks improved decision accuracy for analytical type tasks. Researchers did not investigate the influences of system one intuitive type instructions nor did they investigate how instructions influenced intuitive type decision accuracy. Researchers assumed that decision accuracy confirmed that participants chose analytical thinking style for the analytical type decisions.

Research Question

The research question for this study investigated how instructions for discerning between intuitive and analytical decision tasks influenced a person's choice of system thinking style for resolving analytical and intuitive decision tasks.

- Q1.** What influence, if any, did instructions for discerning between intuitive and analytical decision tasks have on a person's choice of system thinking style for resolving both analytical and intuitive decision tasks?

Hypotheses

- H₀₁.** The intervention group's mean for their choice of system one thinking style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically equal or less than the control group's mean for their choice of thinking style, $H_{01} : \mu_{01} \leq \mu_{a1}$.
- H_{a1}.** The intervention group's mean for their choice of system one thinking

style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically higher than the control group's mean for their choice of thinking style, $H_{01} : \mu_{01} > \mu_{a1}$.

Nature of the Study

The research method of this study was a quantitative, post-test only with control group, experimental design, using multiple treatment groups as defined by (de Vaus, 2001) and specified by Black (1999). This design had two groups which were an intervention group which used instructions on how to determine the differences between intuitive and analytical decisions and a control group which did not use any cognitive instructions. The reason for a quantitative experimental study was because only two groups are needed to test a single independent variable's effect upon a single dependent variables outcome. For both types of decisions, this researcher compared the intervention group's mean score to the control group's mean score for system thinking choice which tested the effectiveness of the instructions for discerning between analytical and intuitive decision tasks. The survey questionnaire provided these instructions as treatment to the intervention group before testing. The control group did not receive these instructions for discernment but received the instructions on how to complete the survey questionnaire before the testing.

These intervention instructions were in addition to the standard instructions which this experimenter provided for participants on the third page of the survey after the informed consent and standard instructions. These intervention instructions for identification of analytical decision tasks were used by Kogler and Kühberger (2007) and specified by Kahneman (2003). These four instructions involved looking for cues;

considering vigilance and accuracy; understanding the amount of time needed for the response; and realizing if taking a second look would help.

All participants completed 10 decision making tasks which consisted of five system one intuitive questions or problems from De Neys and Glumicic (2008), three system two analytical questions or problems from Cognitive Reflection Test (CRT) from Frederick (2005), and two additional system two analytical questions from Böckenholt (2012). This researcher used these decision making tasks to test the participants' choice of thinking style used to resolve the decision tasks. After participants answered each decision making task, the survey asked participants which thinking style they used to resolve that specific decision task. The control group did not receive these additional instructions for discernment but received the regular instructions on how to complete the survey.

Significance of the Study

DST and DPT are helpful scientific theories that offered an explanation of human behavior when making decisions (Evans & Frankish, 2009; Kahneman, 2011). Both theories were clearly stated, were complete, and had high explanatory power when it comes to understanding the two ways in which people think. DST and DPT spawned much research opening new areas for further investigation (Andrews, 2010; De Neys & Glumicic, 2008; Kogler & Kühberger, 2007). Prior research had neglected providing instruction for identifying intuitive decision tasks. Prior research had incorrectly measured accuracy of participants' answers in assessing the effectiveness of system type identification rather than participants' choice of system thinking style (Thompson & Johnson, 2014). This left a gap for research into understanding how instructions for

discerning between analytical and intuitive tasks affected participants' choice of thinking style for their problem resolution.

This paper investigated DST and DPT further, building incrementally upon these previously established theories. There was a gap in literature as intuitive thinking was neglected in decision type identification instructions. Additionally there was a gap in evaluating system type identification based upon choice of system thinking type rather than decision task accuracy (Thompson & Johnson, 2014). This new research needed to investigate how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for their problem resolution. This investigation was not concerned with participants' decision accuracy but rather if they selected the correct thinking style since decision accuracy was based upon many uncontrollable factors.

Definition of Key Terms

Belief-bias. Belief-bias in decision making ignores following a logical process and relies upon what one thinks is true at that moment of decision making (Andrews, 2010). People follow no consistent analytical process but rather base their resolution on feelings.

Biases. Biases are a tendency for people to not think logically but rather respond based upon past feelings, experiences, and impressions (Stupple & Ball, 2008; Stupple et al., 2013; Stupple et al., 2011). There are both intuitive (Andrews, 2010; Kogler & Kühberger, 2007; Stupple et al., 2011) and analytical biases (Goldsmith & Cyboran, 2013; Kahneman, 2011). When people hold their beliefs, intuition, and feelings higher than empirical evidence and rational judgment in decision making, they are demonstrating a system one intuitive bias (Andrews, 2010; Goldsmith & Cyboran, 2013;

Kogler & Kühberger, 2007; Stupple et al., 2011). System two analytical biases are more complicated and demonstrate overvaluing a decision outcome, procrastinating, clue-seeking, over-optimism, false comparisons, and undervaluing costs. Additionally, people consider only available information instead of establishing a criteria and collecting the needed information to make the decision (Barton et al., 2009; Goldsmith & Cyboran, 2013). Due to their biases, people inappropriately override their appropriate system thinking type for their convenience.

Decision Tasks. Decision tasks are questions or problems for participants to answer and resolve which require system one intuitive thinking or system two analytical thinking style for resolution (De Neys & Glumicic, 2008). In this study, a participant's answer to the decision task was not the focus. Instead, it was their choice of system thinking style used for resolving the decision task.

Monitoring. Monitoring in decision making occurs when people allow one of their systems of thinking to observe the other with intentions of overriding. These intentions include forming resolutions; detecting inconsistencies; creating conflict; or identifying anomalies and redirecting thought (De Neys & Glumicic, 2008). System two monitoring occurs when people realize a decision task has conflicting information which requires using analytical thinking over intuitive thinking. System one monitoring occurs when people realize a decision task is too complex or there is not enough time to follow an analytical process and they make a guess at the answer (De Neys & Glumicic, 2008).

Overriding. Overriding in decision making occurs when people allow a system of thinking style to redirect another thinking style (Kogler & Kühberger, 2007; Stupple et al., 2011). System one style overriding occurs when people logically reach a conclusion

that is in conflict with their beliefs and they resolve the decision with their belief ignoring the logical conclusion (De Neys & Glumicic, 2008). System two overriding occurs when people reach a logical conclusion and they ignore their beliefs yielding to empirical evidence in their decision making process (Kogler & Kühberger, 2007).

System One Intuitive Thinking Style. System one intuitive thinking style develops from past experience, impressions, and feelings which a person instantly and impulsively recalls (Kahneman, 2011). People consciously and unconsciously learn from these experiences, impressions, and feelings. People are born with the intuitive thinking ability (Spezio, 2011) which continues to develop and stays with them longer than the analytical thinking style (Boyle et al., 2012).

System Two Analytical Thinking Style. System two analytical thinking style is deliberate thinking which is capable of reasoning that only operates consciously. People use their analytical thinking to monitor their thoughts and they incorporate analytical thinking when decisions are too much for intuitive thinking (Kahneman, 2011). System two analytical thinking develops as the prefrontal cortex (PFC) matures (Fleming & Bartholow, 2013; Ghetti & Angelini, 2008; Wiers et al., 2008).

Instructions for Determining Decision Type. Instructions for determining decision type explain how analytical and intuitive decision types differ (Kogler & Kühberger, 2007). The instructions explain the difference between system one thinking involving intuitive and impulsive thought versus system two thinking which is deliberate requiring analytical thought. The expectation was that through training, participants know when to apply analytical thinking style to analytical decision tasks and intuitive thinking style to intuitive decision tasks. Additionally, by participants knowing which system of

thinking to apply, they chose intuitive thinking style to respond to an intuitive decision task and analytical thinking style to respond to analytical decision tasks.

Summary

Prior research neglected the need for instructions in system one intuitive type decisions and primarily focused on improving only system two analytical thinking. Instructions in identifying decision types had only been tested as it influences system two analytical type decisions, which presented a bias toward analytical thinking and created a conflict when intuitive thinking was all that was required to resolve the problem. Researchers needed to test instructions for discerning intuitive and analytical decision tasks rather than assuming that the system two analytical type was all that was needed. The purpose of this study measured how instructions for discerning analytical and intuitive decision tasks influenced a person's choice of system thinking style for resolving decision tasks.

Chapter 2: Literature Review

The purpose of this study was to gain understanding in how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for their problem resolution. Relevant literature supported the importance of system one intuitive thinking (Fischer & Hommel, 2012; Norris & Epstein, 2011) and expressed different conditions that influenced a person's choice of thinking style for decision resolution. These different influences include time, complexity, self-control, conflict, biases, age, and education (Andrews, 2010; De Neys & Glumicic, 2008; Gibbons et al., 2009; Wills, Pokhrel, Morehouse, & Fenster, 2011; Witteman et al., 2009).

Documentation

This researcher's literature review strategy was to seek out scholarly, scientific, and primary research from peer-reviewed journal articles which were published within the last 5 years. This researcher selected few but very relevant articles or books which were published before 2010. For this study, this researcher included EBSCOhost, ProQuest, SAGE Journals Online, ScienceDirect, SpringerLink, and Wiley Online Library databases in searches. These databases were accessed from the Northcentral University online database library. For the theoretical framework, this researcher sought articles on dual process theory and dual system theory. For intervention instructions, this researcher searched for articles which used primary terms associated with training, teaching, or instruction and secondary terms which included either dual process or dual system thinking. For biases, this researcher searched for articles which used primary terms associated with intuitive biases or analytical biases and secondary terms which included either dual process or dual system thinking. To reduce the number of articles

returned in searches, this researcher included only scholarly, peer-reviewed, and primary research from journals. Together, these relevant articles and books formed the literature review which supported this research providing the reader with a comprehensive perspective. For a listing of search criteria and number of articles found, see Appendix A.

System One

System one thinking is based upon the experiential system developed through conditioning (Norris & Epstein, 2011). There were three forms of conditioning which were classical, operant, and observational learning. Classical conditioning is learning from associated experiences to predict impending outcomes (Gershman & Niv, 2012; Ludvig, Sutton, & Kehoe, 2012). Observational learning involves attentiveness, watching, retaining, imitation, and motivation through the experience of others (Hoover, Giambatista, & Belkin, 2012; Meltzoff, Waismeyer, & Gopnik, 2012; Wesch, Law, & Hall, 2007; Zentall, 2012). Operant conditioning uses reinforcement schedules through negative and positive consequences (Staddon & Cerutti, 2003).

People are more efficient with quick intuitive system one thinking for decisions that require parallel processing or creativity (Fischer & Hommel, 2012; Norris & Epstein, 2011). People are more accurate with slow analytical system two thinking when confronted with conflicting information or complex problems for decision making (De Neys & Glumicic, 2008). System one thinking is automatic and develops from unconscious learning and conditioning as people conduct their lives (Glöckner & Witteman, 2010). System one determines the majority of daily decisions and actions as it is quick and efficient but is not effective in abstract thought. Prior measures for intuitive experiential thinking are ineffective and needed improvement with content validity.

Additionally, researchers need a better understanding of the different facets of experiential thinking (Norris & Epstein, 2011). Fischer and Hommel, Norris and Epstein, De Neys and Glumicic, Glöckner and Witteman, and Norris and Epstein all agree that system one intuitive thinking style is important and necessary. System one thinking style is what everyone uses the most for what they do throughout the day.

Norris and Epstein (2011) focused on cognitive-experiential self-theory which is a sub theory of dual system theory. In the first experiment, researchers measured facets of experiential thinking style using a newly developed scale. In the second experiment, researchers determined validity of this new scale for measuring experiential thinking. In the third study, researchers evaluated relationships between self-ratings and rating by others. They investigated how well this new scale measured experiential thinking style and how closely were relationships between self-ratings and others' ratings for experiential thinking style.

Experimenters conducted three studies to understand experiential versus rational thinking styles associated with creative performance and gender Norris and Epstein (2011). . In the first study, there were 2,245 students who participated with 1475 being female and 770 being male. The second study had 283 students participating with 127 males and 156 females. The majority of the participants were Caucasian with a mean age of 19.1. In the third study, 19 participants had two other people assess their cognitive versus experiential thinking style. In the first study, there were 83% Caucasian with the remaining 17% from various other races of which researchers did not reveal the percentages.

This first study used a survey to measure intuitive thinking along with levels of

emotionality, imagination, and spontaneity (Norris & Epstein, 2011). Other variables included concrete thinking and generalization. This first study used REI and a newly developed scale for measuring cognitive experiential self-theory. In the second study, experimenters used a multitude of instruments which measured visualization, humor, memory, judgment, creativity, intuition, beliefs, social relationship, romantic relationship, self-acceptance, growth, and well-being. Additionally, experimenters measured life meaningfulness, purpose, empathy, self-esteem, and depression. In the third study, researchers used REI and an additional experiential scale. Researchers used the Revised Minnesota Paper Form Board Test of Quasha and Likert (1937) to measure visual recognition. They used Social Intelligence Test of F. A. Moss, Hunt, and Omwake (1949) to measure Sense of humor and memory. They used Design Judgment Test of Graves (1948) to measure judgment. They used the Alternate Uses Task of Christensen, Guilford, Merrifield, and Wilson (1960) to measure creativity. They used the Dyads of Triads task of Bowers, Regehr, Balthazard, and Parker (1990) to measure intuition. They used General Information Form by Epstein and Meier (1989) to measure relationships. They used the Psychological Well-Being Scale from Ryff (1989) to measure self-acceptance and personal growth.

The purpose of this study (Norris & Epstein, 2011) was to understand the positive effects and advantages of system one thinking in the decision making process. Results from this study supported the positive influence of system one thinking upon certain types of decisions requiring creative rather than analytical thinking. System one experiential thinking had a higher positive effect on interpersonal relationships, attitude, and personal improvement than analytical thinking.

In the first study (Norris & Epstein, 2011), results supported that males are more rational than females while females are more intuitive, emotional, and creative than men. In the second study, results supported relationships between experiential thinking and creativity, judgment, humor, and intuition. Experiential thinking style was more closely related with interpersonal relationships than rational thinking style. Experiential thinking style was also more related to growth, self-esteem, acceptance, and life purpose. In the third study, results provided support for validity of new sub scale for experiential thinking style. Additionally, results indicated that self-rating has some bias towards overestimating cognitive thinking over experiential thinking as compared to others-rating.

Norris and Epstein (2011) used Varimax rotation along with their factor analysis in the first study. In the second study, researchers used ANOVA to understand rational and experiential thinking styles. Researchers examined alpha coefficients which supported the reliability of the new scale. This research supported the importance of both thinking styles especially system one with activities that require creativity. Researchers point out the importance of conditioning and learn of the experiential system in quick problem solving and reacting to situations. Additionally, researchers explained how rational system two thinking uses logical principles and evidence in decision making tasks. Analytical system two thinking lacks abstract thought.

Intuitive Heuristics

System one is divergent thinking yielding creativity and providing flexibility (Fischer & Hommel, 2012). System two is convergent thinking yielding analytical thinking and provides a persistent deductive pathway. Analytical thinking tends to be serial and creates limitations in output performance.

Maeyer and Talanquer (2010) examined undergraduate students' heuristic thinking processes when students completed chemistry ranking tasks. They wanted to understand what students would do when they were uncertain of how to respond to decision tasks. Maeyer and Talanquer inquired about the heuristics students used while resolving ranking tasks. They expected students would rely upon intuitive system one thinking and subvert their system two thinking due to time constraints, task complexity, and students' lack of knowledge. They contended that prior research was limited in understanding students' reasoning strategies under these constraints. Maeyer and Talanquer wanted to help students know how to think in more scientific and analytical processes. They wanted to use their research to guide the development of thinking assessment tools for decision making. Their purpose of this study was to examine students' system one and two thinking in resolving ranking tasks.

Maeyer and Talanquer (2010) conducted a mixed method study and used a quantitative experiment and a qualitative interview process. Researchers provided 414 students with questionnaires and interviewed 34 volunteers. The 34 volunteers did not see the ranking questionnaire. Participants answered six questions concerning the arrangement of chemical compounds. Interviewed participants had 60 seconds to answer each question. Interviewers instructed participants to freely speak audibly during testing. Interviewers recorded and transcribed these sessions for analysis. Researchers analyzed student reasoning patterns and coded them for further assessment. They performed circular coding and comparison until the review team reached 90% agreement on coding.

Qualitative threats to validity for this study (Maeyer & Talanquer, 2010) related to transferability, dependability, and confirmability. Students ordered chemistry

substances of which they were familiar from prior training. This scenario drastically limited transferability to other populations. Dependability was at risk due to this specific chemistry context. Because researchers developed their own system of coding and did not present corroborating research, this threatened confirmability. Quantitative threats to validity include internal and external validity (Jackson, 2012). Internal threats for this study Maeyer and Talanquer included nonequivalent control group, testing, instrumentation, experimenter, and subjects effects for validity (Jackson, 2012). Because there was no repeated testing, this study did not have history, maturation, testing, mortality, or attrition threats. The quantitative research threatened internal validity by not providing an equivalent control group and not using measurements which other studies had validated. The quantitative research threatened external validity by using college level chemistry tasks and college chemistry students.

Students consistently used at least one of four different decision strategies as they responded to the chemistry tasks (Maeyer & Talanquer, 2010). These were recognition, representativeness, one-reason, and arbitrary trend strategies. These strategies are system one strategies and not system two. Students tended to make decisions based upon their system one strategies rather than using sound principles of analytical thinking. The constraints of time and lack of information also influenced students to utilize system one thinking over system two. When students applied these system one strategies they significantly decreased their accuracy.

This study (Maeyer & Talanquer, 2010) explicitly stated the need for students to know when to use analytical thinking and provided a process for the students to use. Researchers needed to address qualitative threats of transferability, dependability, and

confirmability by developing a different set of tasks that are not specific to chemistry and college chemistry students. Maeyer and Talanquer needed to address quantitative threats by using measures that other studies have verified. For successful decision making, students need to be able to identify if the decision task requires system one or two thinking. Additionally, students need an understanding of the steps in the system two thinking process.

Böckenholt (2012) focused on dual process theories associated with intuitive and analytical reasoning. He hypothesized that system two analytical responses take more time to develop than system one responses. The amount of time it takes to respond indicates which system of thinking was used. He investigated how item responses identify intuitive or analytical tendencies.

Böckenholt (2012) used three experiments to measure response times to analytical type decision tasks. There were 462 participants in two online studies and 117 participants tested in the laboratory. Researchers used both the CRT and the Morningness-Eveningness Questionnaire (MEQ). Males answered more questions correctly than females. Results supported that system one was important as people used it for depth perception, recognition, association, and implicit learning. People used system two for focus, language, monitoring, and rule-based thinking. System one does not use cognitive capabilities or reasoning but rather beliefs and heuristics.

Böckenholt (2012) pointed out that even when people incorporate system two thinking, their decisions or answers could still be incorrect. Correctness does not determine if people used system one or system two thinking. Response times for system one were shorter than system two as system two took more deliberation. These scholars

claimed that people depend upon system one intuitive thinking because they lack in reasoning skills.

Prior research supported the correlation between higher education and reasoning skills Amsel et al. (2008). The higher training in understanding probability the higher are the analytical capabilities. Researchers should consider other factors when evaluating the complexity of system one versus system two thinking. Some of these factors are testing items, errors, relationships between system one and system two thinking, process characteristics, response times, and confidence levels (Böckenholt, 2012).

Fischer and Hommel (2012) used the Remote Association Task (RAT) to test convergent thinking. They contended that people use cognitive control to regulate multitasking abilities. People should be able to improve multi-tasks efficiency through priming of divergent-thinking. Divergent thinking intervention was accomplished using Alternative Uses Task (AUT). People used divergent-thinking to increase processing quantity by multi-tasking. Their results showed that system one thinking produced more answers faster than system two thinking. System two thinking required a systematic-focus while system one thinking allowed for a higher degree of parallel thought. System two thinking was more costly than system one thinking from purely a production through-put perspective. System one encouraged flexibility and creativity while system two thinking encouraged logical and analytical thought. They recommended future testing which would include a distractive environment or relaxed atmosphere against a distraction controlled environment. Their findings supported the value of both styles of thinking depending on the purpose decision task and the motive of the decision maker which is why this new study included both.

Managerial Behavior and Dual System Thinking

Dual system theory explains the association between managerial behavior and cognitive styles (Cools & Van Den Broeck, 2008). Cools and Van Den Broeck conducted a qualitative study to investigate how managers plan and make decisions. Their proposition was that creative managers use an intuitive approach for decision making. Managers who use evidence and information for planning and decision making follow system two management styles. Executive and managerial coaching has increased without practical cognitive style research (Cools & Van Den Broeck, 2008). Managers' cognitive style or system thinking preferences influences their learning ability, problem solving ability, decision making, communication, interpersonal skills, and creativity. This study addressed how leaders used different thinking styles and affected their behavior.

The purpose of this study (Cools & Van Den Broeck, 2008) was to increase understanding of leaders' cognitive styles and their managerial behavior. Cools and Van Den Broeck contended that there were many quantitative studies but few qualitative studies which provided further insights into different tasks versus people oriented approaches. Their purpose was to understand how different managerial styles influenced interpersonal skills. Researchers collected 553 testimonies from 275 leaders and MBA students along with 278 employees (Cools & Van Den Broeck, 2008). Sample population included 63% men and 37% women with an average age of 30 years old. From 553 testimonies, 100 of the most divergent profiles were considered. From the 100 testimonies, 16 were of the knowing cognitive style, 41 were the planning style, and 43 were the creative style. Researchers performed three stages of analysis to classify data.

Validity threats for a qualitative study are credibility from participants'

perspective, transferability to other populations, dependability, and confirmability through others' corroboration (Trochim & Donnelly, 2008). By selecting the most divergent profiles, Cools and Van Den Broeck (2008) negatively impacted their credibility threat. They negatively affected the threat of transferability by not revealing the gender percentages of the 100 testimonies actually used. They did not reveal the industry of participants nor did they randomly select or assign participants. Cools and Van Den Broeck conceived the categories of knowing, planning, and creating in a prior study (Cools & van den Broeck, 2007) but did not supply corroboration from others which negatively affected the confirmability threat.

Leaders who showed knowing or planning cognitive styles were more analytical while those who showed a creative style were more intuitive and rational (Cools & Van Den Broeck, 2008). Those who were of the knowing style took the longest for decisions while those who were of the creative style took the shortest time. Managers who use a system two thinking style have a tendency to be less friendly and more controlling while managers who use system one thinking are friendlier. Leaders who are task oriented tend to be more analytical system two thinkers, self-controlled, and may appear to be impersonal. Leaders who rely on intuition tend to have strong interpersonal relationships. This research (Cools & Van Den Broeck, 2008) adds credibility to the argument that both intuitive and analytical thinking have value. Researchers of this study needed other researchers to corroborate their measures or they should have used already established measures. They should design a quantitative experiment that would evaluate their support for system one decision making and quantitatively investigate the consequences of leaders not using knowing and planning styles of leadership. Cools and Van Den Broeck

did make a valid case for the value of intuitive system one thinking.

Influences of Memory and Knowledge

Researchers know little about intuitive self-knowledge which is based upon autobiographical memory (Lieberman, Jarcho, & Satpute, 2004). Functional magnetic resonance (fMRI) identifies the neural correlations between intuitive and evidence based self-knowledge systems. Researchers asked how much more do people rely upon their intuition over evidence for areas where they have high and low experience.

Students who were athletes and actors participated in an fMRI experiment (Lieberman et al., 2004). There were 22 participants with 11 being athletes and 11 being actors. Participants responded to words indicating if the words represented them or not. Scanners took fMRI images while participants responded to the words. Athletes and actors responded the same for high and low experience words. When participants responded to low-experience words the area of their brain was most active in their PFC. When participants responded to high-experience words the area of their brain was most active in their x-system regions, basal ganglia.

For high-experience words, participants had activity in the x-system ventromedial prefrontal cortex and low activity in the hippocampus (Lieberman et al., 2004). For low-experience words brain activity was reversed. This supported the use of the intuitive thought for high-experience areas and analytical thought for low-experience areas. When people are familiar with an area due to their experience they rely upon their intuition but when they are placed in an area where they do not have experience they rely upon their analytical thought. This study provides support for the importance of both intuitive and analytical thinking styles as it relates to a person's experience. With more experience less

analytical thinking is required. People utilize more of their self-knowledge for High-experience situations.

Researchers do not fully understand the relationship between people's memory strength and content with respect signal detection theory and the roles these play in confidence, accuracy, and response time (Wixted & Mickes, 2010). Signal detection theory is compatible with dual process theory in remember and know judgments. Intuitive thinking is divided into remember, know, and guess. Researchers performed three experiments. The third experiment tested recognition memory by surveying participants. Participants were to study word targets and lures, memorizing as much as they could. After studying the list, participants were to identify if words presented on a display were from the list.

This study (Wixted & Mickes, 2010) consisted of a quasi-experimental because participants were not randomly assigned and there was no mention of a control group. In the first experiment, there were 25 participants who were University of California students. In the second experiment, there were 31 participants. In the third experiment, there were 35 university students who participated. In the first and second experiment, researchers created a word pool based upon MRC Psycholinguistic Database. The difference between experiment one and two was that a source test was presenting after the targets and lures testing. In the third experiment participants had five options to choose from which were remembering color, location, color and location, knowing, and guessing.

Many assumptions are made in this study (Wixted & Mickes, 2010) as there were no control groups used with all three experiments. All three experiments did not occur at

the same time. Researchers compared and contrasted all three experiments against each other to understand results but the factor of time and lack of random assignment or selection threatened validity. Authors encouraged further study into interactions between single and dual process theories in explaining recognition memory judgments.

Influences of Training and Instruction Intervention

Training and instruction are common interventions (Klossek, Yu, & Dickinson, 2011; Schoenmakers et al., 2010). Klossek et al. (2011) claim that most choice training studies have been performed on rodents and adult humans. Studies with children in response outcomes is unknown for training and learning. Klossek et al. (2011) asked the question on how choice as compared to single action training affect outcome values? Research is also relatively new in the use of attention bias reduction training which encourages people to review all possible options for modifying their behavior (Schoenmakers et al., 2010). The few studies on the influences of attention bias reduction focused on other behavior than alcohol addiction. Klossek et al. and Schoenmakers et al. support the use of training and instruction as a form of intervention.

Once parents provided their written consent for their children to participate, Klossek et al. (2011) divided the children into two groups which were a single action and choice training group. Researchers provided training for the children, gave them a warmup trial, and then tested the children. This was a quantitative study using intervention groups. One group was single action while the other was a choice group. Experimenters trained both groups with no variation in training. Single action participants completed 20 training sets. Once participants completed all the trials, researchers performed the devaluation analysis and extinction testing. There were a total of 42

preschool children who participated in this study from Cambridge in the United Kingdom area. They were from 3 to 4 years old with 24 boys and 18 girls. The independent variable was the two groups of single-action and choice-action. The dependent variable was the value of outcome.

This study (Klossek et al., 2011) used an on-line test for the children which included cartoon videos as rewards. The on-line test recorded the children's presses on the touch screen. The single action group scored higher responses than the choice group. Analysts use mixed ANOVA to evaluate participant responses. Their findings were in-line with other research and confirmed that when people have a choice of different actions they are more goal oriented towards controlling their actions. Generalization of this study was at risk to other populations other than children because this research was designed specifically for children and their thinking levels. The value of this study was that researchers used training methods to influence participants' choice and response outcomes.

Schoenmakers et al. (2010) hypothesized that attention bias reduction training would reduce the influence of alcohol cues. The research question was on how attention bias reduction training affects alcohol addiction in heavy drinkers. This quantitative study (Schoenmakers et al., 2010) used an experiment with two groups consisting of an intervention and control group. The intervention group attended five sessions of attention bias reduction training while the control group received other training tasks. This was a pre-test, intervention, post-test, and 3 month follow up testing design. There were 43 participants with 33 being male and 10 being female. All participants had an alcohol related addiction and volunteered to participate in a study that tested different

intervention procedures. Researchers randomly assigned participants to either the intervention or control group. All participants had similar demographic characteristics.

The independent variables were the group type of either intervention or control (Schoenmakers et al., 2010). Dependent variable was participants' attention bias which researchers measured using a visual-probe task and craving which researchers measured through a questionnaire. Researchers used alcohol related picture sets for the intervention training, visual-probe task for attentional bias, and a questionnaire for measuring cravings. Researchers used ANCOVA to compare pre-test scores and again for post-intervention and testing scores.

Pre-test analysis did not show any differences between control and intervention groups while post-test analysis showed that attentional bias reduction had an effect on resisting alcohol related stimuli (Schoenmakers et al., 2010). There were no significant differences on the effects of attention bias reduction training on cravings. The importance of these studies was the use of training to overcome biases demonstrating the effectiveness of intervention to correct bias behavior which is why this new study used cognitive intervention instruction.

Influences of Time and Complexity

Time constraints and problem complexity also influenced a person to incorporate intuitive thinking over system two (Andrews, 2010; Brisson et al., 2014). As decision tasks become more complicated, people are more inclined to rely upon their beliefs and biases to respond rather than using deductive logic which negatively affects their decision accuracy (Brisson et al., 2014). Dual process theory clarifies that increasing problem complexity increases the need for logical analytical thinking while simple tasks are more

appropriate for intuitive thinking. As problems become more complex people rely more of belief biases especially for those who have low cognitive abilities. Researchers applied dual process theory to conflict problems to determine response times, confidence levels, and logic levels in participants (Stupple et al., 2013). The amount of time people spends on a conflict problem correlates to the accuracy of their decision making. Participants' confidence level is related to their inclination to use either system one intuitive thinking or system two analytical thinking. The greater participants perceive the complexity the more likely they are to resolve the problem using system one intuitive thinking. The greater the complexity the lower the confidence. Andrews, Brisson et al. and Stupple et al. contend that as people perceive there is not enough time to thinking to think through a problem, they are more likely to rely on system one intuitive thinking for the resolution. As decision complexity increases, people also tend to rely more on intuitive thinking rather than thinking harder and longer.

Andrews (2010) investigated how belief-biases of system one thinking and deliberate analytical system two thinking influenced transitive inference. Her hypothesis was that the more difficult and complex the decision situation the more apt people were to rely upon belief-biases overriding system two thinking. She asked four research questions to address this hypothesis. The first related to the correlation between belief-bias, analytical thought, and transitive inference. The second was concerning the effect of order, time, and markedness of terms upon system thinking choice for resolution. The third was the effect of participant conflict identification upon system thinking choice. The fourth was related to participants' ability to override their belief biases when conflict was present or complexity increased.

This study (Andrews, 2010) was a quantitative analysis using three experiments and over 300 participants in total. In the first experiment, researchers used 16 transitive inference test problems to measure acceptance rates and confidence from 19 male and 85 female psychology students. Researchers used the same set of problems in the second experiment but the logical order was changed making the test more complicated. There were 17 male and 83 female university psychology students who participated in the second experiment. The third experiment used the same problem set but time constraints were added. Additionally, researchers incorporated the Culture Fair Test from Cattell and Cattell (1973) to measure fluid intelligence. There were 38 males and 114 females who participated in the third experiment. In all three tests, experimenters measured acceptance rates and confidence. The results supported that an increase in time increased participants' confidence levels and incorporation of system two thinking while a decrease in time negatively impacted accuracy, especially when conflict was present. Lower complexity and consistency increased more analytical thought while conflicting problems in decision tasks increased belief-biases and decreased participants' confidence.

To address influences of time constraints, this new research used a survey tool through (Qualtrics, 2014). There was no time constraints put upon the participant. Additionally, the instructions for the intervention group explained how problem complexity is a contextual cue indicating that the decision task is of system two type structure. For the control group, problem complexity is necessary to measure its influence.

Brisson et al. (2014) conducted three experiments with complexity increasing with each experiment. In the first experiment, researchers tested to what degree of

difficulty participants incorporated more belief bias. There were 191 university students who participated in the first study and used a computer program to record their responses. Experimenters provided students with instructions with the intent to influence analytical thinking for each decision task. In the first experiment, participants were split into two groups with one group receiving limited time to resolve problems while there was no time limit for the other group. This was done to test the impact of time constraints upon participants. In the first study, each participant completed 24 problems with half having conflicting believability and the other consistent believability. Each problem consisted of a rule, observation, and conclusion. Researchers instructed participants to assume the rule to be true and logically deduce the conclusion from the rule and observation. Researchers constructed the second study problems to be slightly more complicated. There were 73 university students who participated in the second study and 196 in the third study. In the first study, independent variables were length of time, difficulty, and problem type while dependent variable was acceptance rates. In the second experiment, independent variables were difficulty and problem type while dependent variable was acceptance rates. In the second study, there were only eight questions with four being conflicting and four being consistent.

In the first study (Brisson et al., 2014), time and difficulty influenced acceptance of the conclusion. In the second and third study, there was a larger variation supporting the effects of complexity on belief biases. Analysts used repeated measures ANOVAs to evaluate data. This study used instructions to influence participants to use their analytical thinking in resolving analytical decision tasks.

A problem with this type of study (Brisson et al., 2014) is that researchers expect

participants to incorporate their beliefs on the rule, maybe the observation, but definitely not on the conclusion. Participants establish a pattern of intuitive thinking for the rule and observation relying upon prior knowledge but then are expected to disregard prior knowledge and think logically to evaluate the conclusion. The instructions tell the participant that they are to believe the rule is true, tell nothing about the observation of being true or false, and then tell participants, they are to logically deduce the conclusion. With this kind of instruction and experiment, decision accuracy could be associated with participants' confusion rather than logic. This study did not base its conclusions on decision accuracy but rather conclusion acceptance. This was why this new study did not evaluate decision accuracy but rather the participants' self-reported choice of system thinking.

Stupple et al. (2013) used a repeated-measures design. Participant resolved invalid versus valid and double negative versus normal affirmative decision tasks. There were 85 participants in the study. Researchers did not capture detailed demographic data. All the participants did not train in reasoning or logic research. Dependent variables included acceptance rates, amount of time to decide, confidence levels, and logic levels. Independent variables were kinds of decisions. Researchers used Authorware software to administer syllogism decision tasks. Participants were to answer syllogism and then provide their confidence level.

Stupple et al. (2013) used a repeated measures ANOVA which provided support to making participants aware of the need to use system two thinking when resolving conflicting complicated problems rather than relying upon their intuition to respond. The more time participants take to respond the more aware they are of the conflict and the

more likely they accept correct conclusions and reject those that are incorrect which is why there were no time limits set in this new study. Problem complexity was also considered in the new study and only decision tasks which had previously been used were included.

Influences of Self-control

When people improve their self-control and self-regulation by utilizing system two thinking they reduce irrational decisions and improve responsible behavior. Wills et al. (2011) tested adolescent usage of alcohol with dual system thinking skills and concluded that impulsive intuitive system thinking leads to higher use of alcohol and tobacco use. Researchers recruited 10th and 11th grade volunteers from the New York area with 52% being female and 48% male. The distribution of ethnicity was 66% Caucasian, 17% Hispanic with the remainder being African American and Asian. Experimenters used a self-report questionnaire to gather data which was approved by the Albert Einstein College of Medicine Institutional Review Board. The good behavioral self-control measure was based upon several previous studies as was the poor behavioral regulation measure. The Good emotional self-control scale came from Kendall and Wilcox (1979) and Zeman, Shipman, and Penza-Clyve (2001). Poor emotional regulation measurement was from Simons and Carey (2002). Academic orientation was from Wills et al. (2006). Negative life events was from Wills et al. (2001). Adolescent's substance use inventory was based upon several previous studies.

Self-control depends on system two thinking with planning ahead, contemplating alternative actions, and predicting consequences for actions (Wills et al., 2011). Poor self-control related to system one being impatient, immediate, and effortless thinking. Self-

control takes a cognitive effort and strategies which were not part of system one thinking. Reflective or deliberate thinking considers future rewards while impulsive or intuitive thinking looks for immediate gratification. Addiction occurs when impulsiveness overtakes reflective thinking.

Their hypothesis (Wills et al., 2011) was that adolescents' substance use correlates to their system two self-control and ability to regulate their behavior. Their research supported that a stronger ability of self-control increased resistance to negative peer pressure. Additionally, a person's ability of forethought helped prepare the person with dealing with problem situations.

Influences of Conflict

People have a tendency to take more time when they recognize conflicting information in the decision making process and incorporate analytical reason-based thinking (Witteman et al., 2009). Making a decision is applying a style of information processing. A person who is characteristically more conscientious demonstrates more of a preference for system two rational thinking. People who are characteristically rational thinkers are less likely to be neurotic, more open to experience, more conscientious, more introverted, and less likely to easily agree.

Witteman et al. (2009) refer to dual cognition theories as two different thinking styles. Their hypothesis was that the higher a person's conscientiousness level, the more inclined a person is to demonstrate a system two thinking preference. The lower a person's conscientiousness level the more inclined to demonstrate a system one thinking preference. They asked to what level is conscientiousness associated with system one or system two thinking styles.

Witteman et al. (2009) conducted three studies with two of them using different populations from different cultural backgrounds while the third study evaluated REI against another system thinking style tool. Experimenters administered a survey questionnaire to participants for both the REI and Big Five assessment. In these studies, there were 774 undergraduate students from Radboud University who participated with 96% being women between 17 and 25 years of age. Experimenters used the Rational-Experiential Inventory (REI) from Pacini and Epstein (1999) to determine participant's thinking style. Researchers also used the Big Five personality measures to evaluate conscientiousness.

Analysts used confirmatory factor analysis to understand relationships between Big Five personality characteristics and system thinking styles (Witteman et al., 2009). Their findings indicated that a person who is characteristically more conscientious demonstrates more a preference for system two rational thinking. People who are characteristically rational thinkers are less likely to be neurotic, more open to experience, more conscientious, more introvert, and less likely to easily agree. This study examines Dual Process Thinking more from a nature rather than nurture position based upon decision making style.

In this new research study, there were no conflicting decision tasks that challenge a person's belief biases. Influences of conflict related to answering a question correctly or incorrectly. This new study differed from this prior study by presenting all believable items. Additionally, this new research measured participants' choice of system type thinking rather than their decision accuracy. Instructions also are associated with identification of intuitive or analytical decision types which removed the threat of

personal biases in decision responses.

Influences of Belief-Bias

People have a tendency to use belief-based system one in their thinking unless they recognize a conflict and incorporate system two thinking (Andrews, 2010; Banks, 2013; Banks & Hope, 2014; De Neys & Glumicic, 2008). Klauer, Beller, and Hütter (2010) showed that more than 80% of the educated adults, who tested, failed in solving reasonably simple decision tasks when conflicting evidence was presented. Dual process theory explains thinking failures by clarifying how people use their bias and intuitive thinking over their logical analytical thinking even when evidence to the contrary is available. Researchers have focused more on accuracy of responses rather than cognitive processes in the dual process theory debate on conflicts between system type choice, monitoring, and decision responses. People have a tendency to seek out evidence that refutes what they do not believe and supports what they believe giving a false sense of analysis and increases belief bias (Stupple et al., 2011). Stupple et al. used dual process theory as their foundation for their study. They hypothesized that participants take a longer time to resolve invalid believable syllogisms than valid or unbelievable syllogisms.

Those who rely on belief biases are less likely to engage in analytical thinking resulting in inaccurate decision making (Pennycook, Cheyne, Koehler, & Fugelsang, 2013). Dual process theory explains the differences in thinking style and how analytical thinking is required to override belief biases from intuitive thinking. Faster response times indicates intuitive thinking style and decreases decision accuracy. Belief bias yields inaccurate results while deliberate analytical thinking overrides belief biases, takes more

time, and increases decision accuracy. People using analytical thinking style make logical decision based upon current evidence set before them while people using intuitive thinking make decisions based upon prior knowledge that is activated by the context of the situation (Klauer et al., 2010). Analytical thinking decides only if the evidence supports the decision while intuitive thinking decides based upon certain situations and then takes action. Prior knowledge supported conditional reasoning. Situations, context, or stimuli triggers prior knowledge. People use intuitive thinking to recall from prior knowledge to give a response. People use analytical thinking where logic is necessary along with abstract content. Intuitive thinking relies upon a knowledge base for accuracy while analytical thinking relies upon logical form.

Klauer et al. (2010) used four experiments to understand how rules impacted participants' decision making. In the first experiment, subject participated in three phases with 1 week between each phase. In the second experiment there were two phases which included one using a rule and the other not. The third experiment had three phases having the first phase without the rule and the second and third phase with a rule. The focus of the third experiment was on rule form. In the first experiment, there were 15 university students with four of them being male and 11 being female that participated. In the second experiment there were 28 participants with 18 female and 10 male university students participating. In the third experiment there were 18 participants with 14 being female. In the fourth experiment there were 13 participants. In the first experiment there were 32 two part problems presenting sufficiency and necessity as either high or low.

This study (Klauer et al., 2010) used ANOVA and repeated measures for analysis. This study contends that people evaluate a decision based upon prior knowledge and then

apply logical correctness based upon form or rule based evidence. System 1 is more efficient than system 2 in speed of processing but may not be more effective in accuracy. This study supported the use of instructions to influence analytical thinking as participants were instructed to read carefully because each problem is unique. Additionally, negative phrases were capitalized for emphasis as these words were often overlooked in pretests. This new study also used instructions as an intervention but no words were emphasized within the word problems.

Banks and Hope (2014) contend that prior research does not explain how both types of system thinking work together for decision task resolution. N200 and P300 amplitude was greater in conflict problems requiring both intuitive and analytical thinking than non-conflict problems requiring only intuitive thinking. Researchers asked how conflict and non-conflict problems influenced intuitive and analytical thinking. Will thinking styles work independent of each other or together for problem resolution?

This study used an electroencephalogram to measure N200 indicating analytical thinking and P300 indicating intuitive or belief type thinking (Banks & Hope, 2014). Researchers administered 192 reasoning tasks which varied from believable versus unbelievable and valid versus invalid. Reasoning tasks or problems consist of at least one premise and one conclusion. Experimenters instructed participants to read each problem, consider each premise as true, and evaluate the conclusion based solely upon the premises. There were 18 participants with 9 being male and 9 being female. Independent variables were problem types of valid believable, invalid believable, valid unbelievable, and invalid unbelievable. Dependent variable was conclusion acceptances.

N200 and P300 amplitude was affected which supported the interaction of logic

analytical thinking and belief intuitive thinking (Banks & Hope, 2014). This was especially true for problems that were conflicting combinations of believable invalid and unbelievable valid. Believable invalid was most significant. Analysts used ANOVA to evaluate participant responses. When problems incorporate conflict between what is logical and what is believable people are more likely to use both intuitive and analytical thinking for problem resolution.

While this study supported both thinking styles working together, ultimately, in the final decision making process, one thinking style overrides or confirms the other for the resolution (De Neys & Glumicic, 2008). Only in the confirmation of both thinking styles can researchers claim that both are working together. This new study recognized the value of both system one and system two thinking styles but also recognized that bias influences one system to incorrectly override the other resulting in error.

Stupple et al. (2011) used a 2 x 2 repeated-measures design. Researchers assigned participants to groups based upon their logic index. There were three groups which were high, medium, and low logic. There were 70 university students and 60 employees who participated with their ages ranging from 18 to 60 years. This population trained in logic and reasoning. Participants solved syllogisms which represented four kinds of decision tasks. These four kinds of decision tasks address combinations of valid versus invalid and believable versus unbelievable problem statements.

Results showed that those in a high-logic group were slower in responding than those in the low or medium logic group (Stupple et al., 2011). Participants took longer to resolve conflicting problems with valid yet unbelievable and invalid yet believable over non-conflicting problems that were invalid yet unbelievable or valid yet believable.

Conflicting problems require participants to take more time to resolve than non-conflicting problems. Researchers used a mixed ANOVA and performed main effect analysis. Participants took the most amount of time to solve invalid but yet believable decision tasks. The higher the logic index the longer the participants took in resolving the invalid but yet believable problems. Those who were in the low-logic group responded the quickest to all kinds of decision tasks. Those who take more time to support and refute an argument apply a higher logic index and avoid analytical biases.

People who allow their prior knowledge or beliefs to influence their analytical type decisions are more prone to error (Trippas, Handley, & Verde, 2013). Dual process theory explains how people use both systems of thinking but rely on intuitive over analytical when decision task complexity increases, time limitations increase, and cognitive capabilities decrease.

Trippas et al. (2013) used two quantitative experiments to understand the effects of complexity, time limits, and cognitive capabilities upon decision accuracy. There were 91 undergraduate students who participated in the first experiment. Researchers provided participants with 64 syllogisms to resolve. Syllogism were valid versus invalid and believable versus unbelievable. There were 85 participants in the second experiment. The first experiment tested complexity while the second experiment tested time limitations and cognitive abilities. The independent variables were complexity level, cognitive capability, and time limitations. The dependent variable was decision accuracy.

Trippas et al. (2013) determined complexity based upon expected number of steps participants would need to take to resolve syllogisms. Time limits were enforced by the computer program for those who were in the time limit group. Control group participants

did not have time limits. Experimenters used an intelligence test to determine cognitive ability.

Results supported that people used their intuition over their analytical reasoning (Trippas et al., 2013). Those who had low cognitive abilities relied more on intuition than those who had high cognitive abilities. The higher the education level the more likely they were to use their analytical thinking style. Those who perceived a lack of time also resorted to their intuition. This study supported that unlimited time and cognitive training increases use in analytical thinking style over intuitive thinking style. Analysts used mixed ANOVA to determine independent variables effect upon decision accuracy.

These syllogisms presented (Trippas et al., 2013) were unfair in that researchers expected participants to rely on beliefs for half of the words of each sentence while they were to ignore beliefs on the other half. Within their example, the first premise has four words of which two are to be ignored (“bird” and “balloons”) and two are on prior knowledge for meaning (“no” and “are”). Participants could have substituted “A” and “B” for “bird” and “balloons” The second premise has two words to be based upon prior knowledge for understanding and two words which were to be ignored as to prior understand. The conclusion in the example cannot be determined as it is possible to be true but there is not enough information to make a conclusion. Participants were expected to understand what valid and invalid mean and determine if the conclusion was valid or invalid. If validity was based upon possibility, it was valid. If validity was based upon enough information, it was invalid. If validity was based upon possibility of being invalid, it was invalid. Researchers threatened their validity of this study when they expected participants to believe half and inhibit beliefs for the other half of each premise

and conclusion. Results may be confounded based upon participants' confusion, reasoning skills, motivation, or memory ability.

Pennycook, Cheyne, Koehler, et al. (2013) asked if people who rely on belief biases are more likely to resolve problems quicker with less accuracy than those who rely on analytical thinking indicated by a longer response time. Researchers administered decision tasks in the form of a survey questionnaire to participants. There were 91 university students who participated with 66 being female. Religiosity scale, Wordsum measures, and syllogistic tasks were the independent variables while believability and accuracy were dependent variables. Researchers used religiosity scales to categorize people into faith based versus atheistic based groups. Researchers used wordsum measures to determine intelligence levels. Mediators were participants' socioeconomic status, ethnicity, and academic demographics. Researchers used syllogistic tasks and response time to test reasoning accuracy and thinking style selection. Researchers assumed that shorter response times indicated system one intuitive thinking while longer response times indicated system two analytical thinking.

Results supported that those who rated higher in religious beliefs were less accurate for conflicting problems than those who rated lower in religious beliefs (Pennycook, Cheyne, Koehler, et al., 2013). Those who rated lower in religious beliefs had longer response times than those who rated higher in religious beliefs. Results supported that those who have stronger religious beliefs rely more on intuitive thinking style than those who are more skeptical. Analysts used regression analysis to evaluate relationships between belief and conflict type tasks. Mediating variables indicated no significance based upon gender, socioeconomic status, or education level.

The importance of this study was the distinction of cognitive ability from style (Pennycook, Cheyne, Koehler, et al., 2013). Personal preference towards a cognitive style is the willingness to utilize analytical thinking for problem resolution even though people may not have the cognitive ability. Decision accuracy measures cognitive ability while selection choice determines cognitive style. Because of this distinction, this new study focused on selection choice of decision style rather than decision making accuracy from cognitive ability.

Pennycook, Cheyne, Koehler, et al. (2013) used syllogism tasks for participants to resolve. Researchers expected participants to use their prior understanding of all terms for the two premises but not on the conclusion which presented a conflict in thinking for participants. Through this testing, researchers expected to be able to determine if participants can separate themselves from their belief biases and draw a conclusion strictly on the assumption that the two premises were true. This type of testing is problematic as participants must not rely upon prior beliefs, understandings, and assumptions to resolve problems correctly but yet rely on new assumptions and understandings presented in two premises before the conclusion. Researchers have inconsistent expectations. For example, the first premise might be “All birds can swim” and the second premise might be “Turkeys are birds” and the conclusion is “Therefore, turkeys can swim”. In the first premise there are four words which must be interpreted, understood, and assumed to be true based upon prior knowledge. Participants were expected to assume “All” means everyone but yet they are to ignore their prior knowledge what a turkey is and that turkeys cannot swim thus raising an inconsistency which researchers used to unfairly draw conclusions. Due to this inconsistency, this new

study did not use syllogism tasks.

Influences of Inhibiting Belief-bias

When analyzing statistical information, people tend to rely upon their beliefs for interpretation and application rather than evaluating the data in an independent manner (Barton et al., 2009). Dual process theory explains how beliefs influence analytical perspectives leading to analytical biases which either seek out supporting data, ignore conflicting data, or skew existing data. Heavy drinkers have low sensitivity to the effects of alcohol, are easily persuaded to drink from cues, and are weak in inhibitory controls (Fleming & Bartholow, 2014). Repeated alcohol use increases impulsive thinking and decreases inhibitions until there are devastating consequences. Dual process theory describes how alcohol cues affect impulsive thinking of wanting a drink while analytical thinking affects realization of consequences, avoidance, and inhibition controls. Barton et al. hypothesized that suppressing belief in decision making which involves reasoning requires more attention focus for resolution. Their question was how belief suppression influences attention in resolving analytical decision tasks.

Barton et al. (2009) conducted two experiments. In the first experiment, attention demands are tested when participants resolve reasoning tasks using their beliefs. In the second experiment, the researcher instructed participants to try not to allow their beliefs to influence their decision making. This was so researchers could measure their levels of attention demands. In both experiments, there were 24 university students from the University of Waterloo who participated in this study. In both experiments, participants resolved eight scenarios in 96 trials with four being believable and four being unbelievable. A computer program presented the trials in random order and tested

participants working memory in a causal decision task. In the second experiment, researchers tested participants' attention when instructed to inhibit their beliefs while they resolved analytical decision tasks. The difference between the first and second experiment was that, in the second experiment only, the researcher instructed participants to avoid past beliefs when selecting options on the questionnaire.

In both experiments (Barton et al., 2009), analysts used repeated measures ANOVA to evaluate participants' response performance. In the first experiment, findings supported that unbelievable scenarios require more working memory than believable. Analytical decision tasks required more focus when beliefs were inhibited. In the second experiment, findings supported that beliefs can be inhibited through instruction but at a cost requiring more attention to decision making. This research supported that people who use the system two analytical thinking style are still susceptible to bias. The contribution of this study was the testing of bias inhibition instructions improving analytical thinking style. This study supported the use of belief inhibiting instructions to influence decision accuracy. In this study, this researcher administered similar discerning instructions to test its influences upon choice of thinking style.

Fleming and Bartholow (2014) hypothesized that drinkers with low sensitivity to the effects of alcohol are more prone to alcohol cues and have less control than those with high sensitivity. School administrators conducted a survey using the Alcohol sensitivity Questionnaire and from this population of 2000 students, randomly selected a sample of 200 students. Fleming and Bartholow conducted a screening process from 200 students providing that they wished to participate in this study. From these interviews, researchers made their final selection of 85 undergraduate students to participate. The school

rewarded those who participated with course credit. These participants who completed self-report questionnaires. The independent variable was high or low alcohol sensitivity while dependent variables were approach bias, Go No-Go accuracy, inhibition accuracy, and typical alcohol usage. Experimenters used the Alcohol Approach Avoidance Task to measure participants' responses of desire to images which sometimes included alcohol. They also used the Go No-Go Task to measure inhibition control. Participants completed an alcohol family history, negative consequences, and typical use questionnaire. Additionally, experimenters used an electroencephalogram during the Cued Go No-Go Task.

Results supported low sensitivity to the effects of alcohol increase the risk of alcohol misuse as it correlates to stronger approach bias and weaker inhibition control than high sensitivity (Fleming & Bartholow, 2014). Analysts used a mixed factor ANOVA to evaluate results. This study encourages more research into inhibition control of approach biases especially for those who have high potential in alcohol misuse. While this new study was not associated with alcohol misuse, it focused on testing inhibiting bias through discerning instructions.

Influences of Analytical Willingness

People need to understand how their willingness versus their ability to engage analytical thinking affects their responses to moral value judgments (Pennycook, Cheyne, Barr, Koehler, & Fugelsang, 2013). Participants' cognitive ability and style have different effects upon their moral judgments. Ability and willingness increase reflection time upon moral judgments while ability increases accuracy.

For their study, Pennycook, Cheyne, Barr, et al. (2013) recruited 570 participants of which only 505 were included due to 65 having missing data or attention issues.

Almost half were female with the mean age of 31 and all were US residence.

Experimenters administered base-rate tasks, CRT, and Wordsum tests to evaluate cognitive willingness and ability. Additionally, participants answered moral judgment, values, religiosity, political ideology, and demographic questions.

Participants had a high correlation between willingness and ability, between belief and moral judgments, and a negative correlation between moral judgments and either willingness or cognitive ability (Pennycook, Cheyne, Barr, et al., 2013). Those who relied more on the intuition rated higher in moral judgments than those who expressed willingness to use analytical thinking style. Analysts used multiple regression analysis in predicting relationships between analytical thinking ability and style with moral judgments, values, and beliefs. The importance of this study to this author's research was that Pennycook et al. recognizes the importance of testing for analytical willingness over testing for correct answers. Correct answers indicated ability while incorrect answers was not an indication that participants were unwilling to think analytically. Willingness was demonstrated in the increase of time participants took to reflect on analytical questions and with their express thinking style preferences.

Influences of Rethinking

When it comes to decision making, people tend to evaluate their thinking style based upon the false probability of being successful over the true value of the outcome over time (Thompson, Evans, & Campbell, 2013). Impatient leaders expect and reward people for immediate answers over well thought through responses. When people

perceive a decision task to be easy to resolve, they use intuitive thinking over analytical thinking. When people quickly decide upon something, they are more confident in their answer and less likely to change their answer. Researchers wanted to know why people who have high cognitive abilities to think analytically do not use it for certain decision tasks.

Experimenters instructed participants to quickly respond to a decision task, record their confidence level, and then rethink to answers again to see if they would change their mind (Thompson, Evans, et al., 2013). This study used a quantitative experimental between-subjects design. There were 208 people who participated in this study. Experimenters split the participants into either a control group or intervention group. Out of the 208, 150 were women and the overall mean age was 19.9 years. Out of the 208, 185 were university students. The control group provided a single answer while the intervention group provided two answers. The intervention group would quickly respond under a time limit, rate their confidence level, and then respond again with no limit on time. Independent variable was first versus second response while dependent variables were confidence level, length of time, number of changes, and decision accuracy. Researchers used Wason's selection decision tasks that consist of a deck of cards to test matching bias. They formed eight problem statements with problems consisting of a rule, consequence, and status of true or false which formed 32 decision task combinations. Experimenters executed the testing for participants through a computer program. Experimenters instructed intervention participants that their second answer was final and that participants should ensure they answered correctly. The computer system displayed a message for participants indicating when it was the first answer as opposed to the final

answer.

Analysts used ANOVA to evaluate the data (Thompson, Evans, et al., 2013). Findings supported their hypothesis that when people give quick answers, they are more confident, and less likely to change their mind. When people respond using their beliefs rather than logic, they are more confident in their answers, and when given the opportunity to change their mind, they are would rather not take the time. This would indicate that to influence analytical thinking, people need to perceive they have adequate time to respond and are expected to give one final answer. Once an answer is given, people are less apt to change their answer. In this study, researchers did not use decision accuracy to determine which thinking style participants chose to use. Instead, researchers used time taken to respond and number of changes. This provided support in this new study not to use decision accuracy for identifying choice of system thinking.

Different levels of cognitive skills affect choice of system thinking even when people are instructed to rethink their answers (Thompson & Johnson, 2014). Dual process theory explains how quick first responses are usually intuitive while rethinking leads to more analytical thought. Additionally, dual process theory explains how those who have higher cognitive skills are more apt to apply analytical thinking when instructed to rethink a decision task. Researchers predicted that participants who scored higher on their Intelligence Quotient (IQ) test would take longer in time for rethinking than those who scored lower on their IQ test. Those who scored higher on their IQ test would change their answers more frequently than those with lower IQ scores. Differences in IQ scores predicted how early decision making occurred and changes were made during rethinking process. Researchers asked how much rethinking responses influenced

changing it and how rethinking affected accuracy for analytical decision tasks.

Researchers (Thompson & Johnson, 2014) administered an IQ test and a set of decision tasks to participants. Participants answered the decision task two times. The first time participants were instructed to quickly answer questions based upon their first response. The second time participants took the test researchers instructed them to rethink their answers taking as much time as needed to see if they wished to change them. There were 117 participants in this study of which researchers used only 103 responses. Out of the 103, 61 were female and the mean age was 22.4 years. Most of the participants were psychology students at the University of Saskatchewan (N=69). Researchers randomly assigned participants to one group out of four. Independent variables were quick first response versus second response and IQ scores. Dependent variables were problem response scores, response times, number of changed answers, and confidence ratings per problem type. Experimenters administered 24 reasoning tasks through an interactive computer program. Reasoning tasks included neglect problems, denominator neglect problems, causal reasoning, and belief bias syllogisms. Problems were split between those with conflicting and consistent information. Participants also completed an IQ test and a system thinking style preference test. To divide the IQ scores, analysts used a medium split. Additionally, participants indicated how confident they were in their answers after both sets of testing.

For conflicting problem tasks, confidence was lower than non-conflicting tasks (Thompson & Johnson, 2014). For changes in answers, conflicting problems were higher than non-conflicting problems. For the first response, there was no significant difference between conflicting and non-conflicting problems in response time. For the second

rethinking response, conflicting problems took longer than non-conflicting problems in response time. Analysts used mixed ANOVAs to evaluate data. The contribution of this study was the process researchers took to instruct participants to rethink their answers. This instruction to see if a second look would help is one of the four which Kahneman (2011) proposed and was used in this new study. Findings indicated that a second look or rethinking the problem did encourage high IQ participants to take more time and change more answers than the lower scoring IQ participants (Thompson & Johnson, 2014). Both groups were just as confident in their answers as the other but there was no difference over all in their accuracy for certain types of questions. This would support that a second look incorporated more analytical thinking for analytical tasks but not higher decision accuracy. This study provided support for this new study and focused on choice of system thinking rather than decision accuracy as decision accuracy may be more related to technical skill and knowledge.

Influences of Habits and Intentions

Because strong habits override positive intentions, researchers need to know more about developing positive habits that support positive intentions (Rebar, Elavsky, Maher, Doerksen, & Conroy, 2014). People must increase their intentional strength to overcome their habitual behavior. Dual process theory explains how habits are based upon intuitive and impulsive thinking while intentions are based upon deliberate and analytical thinking. Researchers investigated how intentions working with positive habits influenced behavior.

Rebar et al. (2014) conducted investigative research on physically active people using a survey, daily self-reporting questionnaire, and accelerometers. There were 76

women and 52 men who were university students that participated in this study with the mean age of 21 years. Participants completed a survey questionnaire about their habit strength and physical exercise. Additionally, researchers equipped participants with an accelerometer to track their physical activity. The experiment lasted 13 days. Participants recorded their behavior, habits, and intentions daily into a web application. Independent variables were habit strength and intention strength levels. Dependent variable was daily physical activity measure in number of hours.

Results indicated that to maintain a positive behavior, both habits and intentions improve success. When habits are weak, stronger intentions are needed. When intentions are weak, stronger habits are needed. Analysts used hierarchical linear regression model to evaluate the interactions of habit and intention and these effects upon physical activity. When people learn to use positive impulsive habitual thinking with positive deliberate intentional thinking, they are more successful in positive behaviors. The value of this study was the support that both intuitive and analytical thinking styles were important for successful decision making which was the contention of the new study.

Influences of Monitoring and Overriding

Few research studies explain how the mind monitors and controls the decision making process (Thompson, Turner, et al., 2013). Dual process theory holds the position that intuition produces a quick answer while reasoning takes longer. When people quickly perceive they understand decision tasks they more often incorporate intuitive thinking over analytical thinking while if the perception is that the decision task is complicated, those with higher cognitive abilities take more time to resolve it. Those with lower cognitive abilities override their analytical thinking and fall back on their intuitive

thinking due to the problem complexity. Researchers hypothesized that participants' speed in which they resolve decision task correlates to confidence in correct resolution. Participants' speed of understanding a decision task correlates to their confidence in correct resolution.

This study conducted 4 experiments (Thompson, Turner, et al., 2013). The first experiment had 108 university students participate with 76% being female. The second experiment had 72 university students participate with 78% being female. The third experiment had 150 university students participate with 28% being female. The fourth group had 142 university students participate. Independent variables were time restrictions, reading difficulty levels, and problem types. Dependent variables were time taken, confidence levels, number of changes, and decision accuracy. The first experiment used 16 conditional statements which resulted in 64 decision tasks. Researchers formed the conditional statements into combinations of believable versus unbelievable and valid versus invalid problems. Experimenters manipulated the font size, color, and shape to increase difficulty for 30 participants to test their perception fluency. In all experiments, participants were instructed to give a quick answer and then provide their confidence level for correctness. Once the system recorded the confidence levels, the participants were allowed to rethink their answer and correct it if they so desired with no time limits. The system encouraged participants to reevaluate their answer taking as much time as needed for resolution.

The second experiment used 18 base rate probability problems which split into three types of decision tasks (Thompson, Turner, et al., 2013). The first type was incongruent which challenged stereotypical biases. The second type was congruent which

aligned with stereotypical biases. The third was neutral which avoided stereotypical biases. The third experiment was similar to the second with the exception that SAT scores were used to determine cognitive ability and participants were split into four groups. There were two groups that tested perceptual fluency and two groups that tested confidence levels. In the fourth experiment, researchers randomly assigned participants to either the difficult font or the easy font group. In addition to prior experiments, these participants completed a 41 item open-minded questionnaire.

Finding from Thompson, Turner, et al. (2013) showed that participants were more confident with answers that quickly come to mind than they were when they take longer time thinking. Even with taking longer, participants' decision accuracy remained the same for those with low cognitive capabilities. Analysts used mixed ANOVAs to evaluate data for the first three experiments but for the fourth analysts used ANCOVA to evaluate relationships between cognitive capabilities and open-minded scores. Findings supported that increasing the difficulty to read due to the font size or shape increased decision accuracy for those who had high analytical skills and abilities.

This study (Thompson, Turner, et al., 2013) used the font size and shape as an intervention to affect decision accuracy for analytical problems. Only those who had high cognitive abilities improved their decision accuracy which is why decision accuracy is not a good measure for determining a participants' choice of analytical thinking style. Even when people incorporate analytical thinking for problem resolution they still may answer the problem incorrectly. In this study, both high and low cognitive participants took more time for problems that were difficult to read due to the font. Both high and low cognitive participants increased the number of changes after rethinking but only high

cognitive participants improved their accuracy. This new study used the same Cognitive Reflexive Test but evaluated choice of system thinking instead of decision accuracy. This study only focused on analytical thinking while this new study included the same number of intuitive tasks in testing discernment instructions.

De Neys and Glumicic (2008) focused their research problem around conflicting theories as to whether people use system two to monitor their system one responses with logical analysis or use system one to monitor system two responses with their internal feelings. They asked the question, when people realize they have conflicting information, which system is monitoring which response? People may not realize that either system is monitoring the other. They also may not realize which system responded when they change their mind. A person's ability to successfully resolve conflict is dependent upon knowing which system of thinking needs to respond.

De Neys and Glumicic (2008) conducted two experiments with one using base rate problems and speaking aloud to test if system two thinking would override system one. This was due to the conflicting information in the decision task with the base rate data. The second experiment tested response time predicting a longer time would be taken for incongruent problems over congruent control questions when participants identified conflict with the incongruent problem.

Results from this study (De Neys & Glumicic, 2008) supported that accuracy on incongruent problems was lower than congruent control questions. When people were presented with conflicting information they relied more on system one thinking. Those who noticed the conflicting base rates, answered the questions correctly indicating they used system two thinking in their responses. Those who took a longer time in answering

questions were also more accurate. People favored using a heuristic response from system one thinking due to inhibition control failure from system two. This occurred even when a conflict in information was presented and was identified. People trusted their intuition over their analytical thinking. Personal biases were stronger influencers than logical thinking.

De Neys and Glumicic (2008) encouraged training for people to help them override their temptation of providing a system one response when system two information and adequate time was available for logical analysis. People need help in identifying when adequate information is available and that they do not need to rely upon intuitive thinking. There is still a debate with whether system one and system two operate in parallel, serial, or hybrid two-stage model; which system acts first; and which system overrides the other. Additionally, the question remains as to whether training or tutoring can improve system thinking choice or accuracy. According to De Neys and Glumicic (2008), many other questions remain unanswered about system two monitoring system one with analytical thinking due to the overriding effects of intuitive bias.

De Neys and Glumicic (2008) call for further analysis in understanding the shallow monitoring of system two over system one and if a deeper system two analysis can override system one bias. This is because, even though people recognize conflicting information, they still discard the facts and rely on their own personal bias. This new research further investigated how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for their problem resolution. This study predicts these instructions assist people in identifying decision task types based upon available contextual cues. This indicated that participants relied upon their

intuitive thinking or incorporated more analytical thinking.

Influences of Involvement

Advertisers have focused their attention to customer demographics rather than customers' level of involvement missing targeted sales opportunities based upon a dual process model (Wonseok, Yong Jae, & Stepchenkova, 2014). Dual process theory explains how the more involvement a person has the more effortful and analytic their thinking becomes while those who are not involved are more inclined to an emotional intuitive response. Wonseok et al. claimed that participants' involvement or lack thereof influenced their choice of system thinking style. They questioned how if any will participants' involvement, influence their thinking style.

Experimenters (Wonseok et al., 2014) used a between-groups design testing utilitarian versus hedonic events along with informational versus emotional messages based upon situational involvement being high or low. Participants used a computer laboratory for the testing completing a questionnaire through a web based program. Researchers recruited students from sports related classes to take part in the experiment. Researchers randomly assigned 215 students to one group out of eight different scenarios. Each group consisted of 25 to 30 students. Independent variables were the two different events, type of message, and situational involvement. Researchers tried to account for confounds of prior knowledge and participants' attitude by using fictitious names for the events. Internal validity is at risk with so many factors to control being that there were two events used with testing that are unrelated. The dependent variable was the participants' attitude towards the advertisement measured using a 7-point Likert scale.

Analysts (Wonseok et al., 2014) used ANOVA and Cronbach's alpha to evaluate

relationships between informational versus emotional messages and high versus low participant involvement as it influenced participants' attitudes towards the advertisement and event. Their findings indicated that advertisers' informational messages have a stronger appeal on sporting event customer who are more involved with the sports than those who lack involvement. Advertisers' emotional appeals have a stronger effect on customers who lack involvement in sporting events more than those who are engaged.

Researchers should investigate if this is generalizable to managers and their involvement in gathering information for decision making. They should ask what type and how much, if any, involvement in data gathering influences their thinking style in decision making. The implication of Wonseok et al. (2014) upon this researcher's study was that those participants who are more involved in decision making activities may be more prone to analytical thinking style when testing. Because this study used a control group, this effect was minimized as both groups were assumed to be statistically equivalent with random selection and assignment (Black, 1999).

Influences of Relationship and Coordination

Leaders and subordinates relationships are at different levels ranging from friendly to contractually which has impacted individual and group performance (Li & Liao, 2014). Dual process theory explains subordinates perceive leads as friendly when they have a more intuitive thinking style while those who are more structured and objective have a more analytical thinking style. This study included two other theories. The first was role engagement theory which evaluates subordinate engagement while the second was role system theory which evaluates subordinates coordination. The quality of relationships between leaders and subordinates influences group performance.

Employees' engagement influences the quality of leader and subordinate relationships impacting performance. The quality of relationship interaction impacts team coordination. The more isolated a team is the worse their coordination. Group coordination improves group performance and quality of leadership member interaction. Researchers asked how the quality of leader and subordinate relationships impact group performance.

Li and Liao (2014) administered three surveys and evaluated archive data for employees at a Chinese bank. There were 495 participants who completed the first survey, 433 who completed the second follow up survey, and 386 who completed the third survey. Participants were from China and their mean age was 32 with almost equal percentages of females and males contributing. Of this sample, 86% were college graduates. Researchers measured leadership and subordinate interaction using a 7-point Likert scale with 12 items, employee engagement using a 13 item scale, coordination with a three item scale, and job performance with archival data.

Analysts used logistic regression to detect bias from responses (Li & Liao, 2014). Analysts dropped out surveys with either missing data or team groups who did not have enough participation. Analysts used confirmatory factor analyses to evaluate relationship quality, engagement, coordination and job performance. Findings supported that the higher the quality in leadership and subordinate relationship and the higher engagement the better the individual job performance. As quality of leadership and subordinate interaction improved so did the groups coordination. As the group coordination improved so did the groups' performance. This study supported leaders using both a friendly intuitive thinking style to build relationship and an analytical thinking style to support

coordination of roles. This supported the need for both thinking styles in decision making as this study proposes. This prior research demonstrates the importance of system one thinking style within the work place and by members of management.

Decision Making and Diversification Bias

According to Kogler and Kühberger (2007), people compulsively use a lottery strategy to predict outcomes for probabilistic tasks. Even though a probability maximization rule strategy would improve accuracy, human preferences are to use diversification biases. Their perception is that a matching strategy offers more opportunities of success even though there is little probability while probability maximization strategy guarantees error. This study used dual cognitive process theories and claimed that system one thinking influenced diversification bias while system two thinking was analytical and capable of correcting system one biases.

Kogler and Kühberger (2007) hypothesized that people have a diversification bias due to system one thinking. Researchers wanted to know if their instructions in system two identification would help participants take corrective actions, appropriately select system two, and improve accuracy. Kogler and Kühberger claimed that literature was lacking in establishing corrective actions for system thinking choice. Kogler and Kühberger experimented with system two type identification. They provided participants with system two instructions identifying contextual cues to influence their system thinking choice. The purpose of this study was to investigate how system two type identification influences subjects' system thinking choice.

Researchers used card task probability matching to measure responses (Kogler & Kühberger, 2007). Experimenters provided instructions for system two corrective actions

of identifying cues, monitoring, and statistical training. The research design was a quantitative posttest only experiment with a control group. Researchers recruited 97 university undergraduate students experienced in statistical probability. The independent variable was group type of either treatment control group. Selection of either maximizing or matching was the dependent variable which researchers determined based upon card-task selection. Kogler and Kühberger used a control group in their experiment but did not explain how they did selection or assignment to address the nonequivalent control group threat. Having a control group did address the history, maturation, testing, and instrumentation effect. Scores and groups were equally analyzed which addressed the regression to the means threat. Researchers used measuring tools that other research had already established and reduced floor and ceiling effects. Researchers tested subjects within a short time period with no retesting and reduced mortality and diffusion of treatment threats. They did not attempt to address external validity as their experiment focused upon students with a statistical background and the decision tasks were statistically oriented. Participants were 78% female but the study did not have an analysis on gender or other demographics which opens scrutiny due to potential demographic confounds.

Results supported that strengthening system two identification of contextual cue, bias correction, and monitoring of the decision process improved decision accuracy for probabilistic tasks (Kogler & Kühberger, 2007). By participants perceiving the probabilistic task as an analytical process rather than a lottery, they were able to overcome their diversification biases and improve their accuracy. People have a tendency to trust system one plausible judgment over system two effortful thinking. This is because

results come quicker and require less effort. People doubt their plausible judgment when system two thinking is monitoring their system one responses. People who do value statistical analysis are more likely to fail when presented with probabilistic decision tasks. The significance of this study was in its establishment of an intervention of system two type identification to correct diversification biases. Because this research focused upon analytical thinking, this researcher tested the intervention of system one type identification to see if it improved intuitive decision tasks.

Influences of Age and Education

Issues arise when people have decisions that require analytical thought and control and they incorporate their intuitive and impulsive thinking over system two making irrational decisions (Gibbons et al., 2009). As adults decline cognitively, they experience serious consequences and make poor decisions (Boyle et al., 2012). Adults' poor decision making is related to their cognitive decline as they age.

According to Amsel et al. (2008), a person's age and education increases his or her ability to control and utilize analytical thinking over system one. They asked how a person resolves the system thinking choice when conflicting information has been presented. Their research involved a quantitative experiment using within subjects and a second experiment using participants randomly blocked into different ratio-bias tasks to understand different aspects of either nature, development, or stability in system thinking. The first experiment required adult subjects and enlisted 352 university psychology students. Of these participants, 67% were female and the mean age of 20.53 years. University student volunteers received a research credit for their participation. For the first study, researchers developed their own questionnaire that was patterned after

previous studies. The second experiment engaged 304 participants from middle-schools with 120 being female and 184 male. Researchers used a similar questionnaire and tasks as was used in the first study. Results supported that a higher metacognitive status decreased the number of preference selections indicating better system two thinking selection over system one.

In this research, this researcher captured demographic information. This researcher evaluated the demographic information of both groups to understand if it had an influence. Qualtrics randomly selected and assigned participants to control and intervention groups which ensured statistical equivalence. This reduced the influence of age and educational levels. Results disclosed a demographic summary of both groups along with the total of participants.

Boyle et al. (2012) perform cognitive evaluations and administered surveys on participants from the Memory and Aging Project. There were 420 people from 40 different senior homes who participated of which none of them were suffering from dementia at the time of this study. They used the Decision Making Competence Assessment Tool for decision making assessment, a 5-item scam susceptibility test, and 21 other tests for cognition assessment. Results supported that adults' decision making skills decline as their cognitive abilities decline with age. Researchers compared covariates associate with decision making and with scam susceptibility tests, calculated slope for decline, fit the model to a linear model, and used least squares regressions for decision making and scam susceptibility predicting. There was a relationship between cognitive decline and decision making over a 5 year period and there was a relationship between susceptibility to scams.

Other Fields of Research

Marketing researchers have used dual system thinking to explain and manipulate people's buying patterns and influence their choices (Goldsmith & Cyboran, 2013; Mao & Oppewal, 2012). Many have contended that system thinking choice may be influenced through training with the purpose of improving decision making by using the appropriate method for different types of decisions (De Neys & Glumicic, 2008; Klein & Weiss, 2007; Kogler & Kühberger, 2007; Prowse Turner & Thompson, 2009). As such, this was the purpose of this new research in understanding how instructions which determined differences between analytical and intuitive decision tasks influenced a person's choice of system thinking style for resolving decision tasks.

Neuroscience research supported dual system thinking by providing images of brains that showed a shift of blood flow in regions of people's brain based upon their system thinking choice in decision making (Bechara, Damasio, Damasio, & Anderson, 1994; Cohen, Braver, & Brown, 2002; Diekhof, Kaps, Falkai, & Gruber, 2012; Durstewitz & Seamans, 2008; Houdé et al., 2000; Kalish & Dunn, 2012; Spezio, 2011; Wright, 2012). Bechara et al. (1994) examined 20 patients who had prefrontal cortex damage and conducted three different experiments with subsets of the 20. Additionally, 44 control subjects also participated in the experiments. Researchers were investigating the effects of prefrontal cortex damage upon decision making tasks. Results supported that those who had experienced damage to the prefrontal cortex area did not take potential consequences of their actions into consideration in their decision making. These subjects were only interested in immediate rewards. Findings from this study supported the PFC as the section of the brain which provided system two thinking.

Researchers generally agreed that people are born capable of system one thinking, that it is natural, and impulsive while system two thinking begins as the prefrontal cortex matures (Ghetti & Angelini, 2008; Spezio, 2011; Steinberg, 2008; Wiers et al., 2008; Wright, 2012). Ghetti and Angelini (2008) examined memory recall and recognition with children and adolescents in two experiments. The first experiment had 117 children, who ranged in ages of 6 to 18 years. The second experiment had 56 children participate who ranged in ages of 6 to 10 years. Results supported the increase in recollection and recognition development as children increase in age.

Wesley, Hanlon, and Porrino (2011), in their research with chronic marijuana users, supported the activity of the prefrontal cortex for non-marijuana users while those who were chronic users lacked prefrontal cortex activity. Those who were non-users were able to successfully incorporate system two thinking and control while those who were users struggled with system two thinking and relied upon intuitive and impulsive thought. This study included 32 participants, split into two groups and gathered data from Functional Magnetic Resonance Imaging (fMRI) while the participants completed the Iowa Gambling Task (IGT).

Conflicts and Different Perspectives

Researchers have extolled and disputed the advantages and disadvantages of both systems of thinking and identified vulnerabilities on both sides to biases that may lead to suboptimal decision making (Banks, 2013; Davies & Turnbull, 2011; Fleming & Bartholow, 2013; Kogler & Kühberger, 2007; Stuppel et al., 2013; Ward, 2000). Many have seen a conflict and competition in the decision process between system one intuitive and system two analytical with each system struggling for control in the minds of people

(Allen, 2011). Some see that both systems are complementary, supporting the weaknesses of the other system. System one weaknesses are the belief-biases and impulsiveness (Andrews, 2010) while system two weaknesses are rigidity, slow responsiveness, and lack of creativity (Norris & Epstein, 2011).

Summary

There are many negative and positive influences in decision making. While time constraints and complexity negatively influenced choice of thinking style due to biases, researchers have investigated rethinking, engagement, and instructions as positive influences. Barton et al. (2009), Kogler and Kühberger (2007), and De Neys and Glumicic (2008) studies were examples of how dual system theory applied to decision making. While their research tried to influence system two choice to improve accuracy, there was still a gap in research in understanding how system one instruction influences system thinking choice for intuitive and analytical decision tasks. The purpose of this study was to understand how instructions which distinguished differences between analytical and intuitive decision tasks influenced a person's choice of system thinking style for resolving decision tasks.

Chapter 3: Research Method

Prior research focused on instructions for analytical decision tasks and ignored intuitive type decisions (De Neys & Glumicic, 2008; Stupple et al., 2013). People are prone to errors from both intuitive and analytical biases (Andrews, 2010; Goldsmith & Cyboran, 2013; Kahneman, 2011; Kogler & Kühberger, 2007; Stupple et al., 2011). The problem addressed was in understanding how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for their problem resolution. The purpose of this quantitative study was to gain understanding in how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for their problem resolution.

Research Question

The research question for this study investigated how instructions for discerning between intuitive and analytical decision tasks influenced a person's choice of system thinking style for resolving analytical and intuitive decision tasks.

- Q1.** What influence, if any, would instructions for discerning between intuitive and analytical decision tasks have on a person's choice of system thinking style for resolving both analytical and intuitive decision tasks?

Hypotheses

- H₀₁.** The intervention group's mean for their choice of system one thinking style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically equal or less than the control group's mean for their choice thinking style, $H_{01} : \mu_{01} \leq \mu_{a1}$.
- H_{a1}.** The intervention group's mean for their choice of system one thinking

style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically higher than the control group's mean for their choice of thinking style, $H_{01} : \mu_{01} > \mu_{a1}$.

Research Methods and Design(s)

The research method of this study was a quantitative, post-test only with control group, experimental design, using a treatment groups as defined by (de Vaus, 2001) and specified by Black (1999). This design had two groups which were an intervention group receiving instructions on how to determine the differences between intuitive and analytical decisions and a control group which receives no instructions. The reason for a quantitative experimental study was because only two groups are needed to test a single independent variable's effect upon a single dependent variables outcome. For both types of decisions, the intervention group's choice mean was compared to the control group's choice mean which tested the effectiveness of the instructions for discerning between analytical and intuitive decision tasks. The survey questionnaire provided these instructions as treatment to the intervention group before testing (See Appendix B). The control group did not receive these instructions for discernment but received the standard survey instructions on how to complete the survey questionnaire before the testing. The standard set of instructions is on page 2 of the survey after the informed consent page (See Appendix B).

These intervention instructions were in addition to the standard instructions which this experimenter provided participants on the third survey page after the informed consent and standard set of instructions. These intervention instructions for identification of analytical decision tasks were used by Kogler and Kühberger (2007) and specified by

Kahneman (2003) These four instructions involved looking for cues; considering vigilance and accuracy; understanding the amount of time needed for the response; and realizing if taking a second look would help.

Participants completed 10 decision making tasks which consist of five system one intuitive questions or problems from De Neys and Glumicic (2008), three system two analytical questions or problems from Cognitive Reflection Test (CRT) from Frederick (2005), and two additional system two analytical questions from Böckenholt (2012). This researcher used these decision making tasks to test the participants' choice of thinking style used to resolve the decision tasks. After participants answered each decision making task, the survey asked participants which thinking style they used to resolve that specific decision task. The control group did not receive these additional instructions for discernment but received the regular instructions on how to complete the survey.

One alternative to this design would be to provide an additional pre-test to first understand the subjects' system thinking style preference. This would help in establishing if either the control group or the treatment group were more likely to use system one intuitive or system two analytical thinking style without the intervention. Because participants were randomly assigned, a pretest was not necessary with both the treatment group and the control group was probabilistically equivalent (Trochim & Donnelly, 2008).

Another alternative method would be to have an additional form of treatment to function like a placebo. This would address the participant expectation effect and could be in the form of classical conditioning or practice problems providing pre-training before the testing (Jackson, 2012). The reason for not including this alternative was that

there was not enough support from literature for the effects of conditioning in system thinking choice. Because this researcher used a method which randomly assigns all participants into either a treatment group or a control group, an additional placebo group was not required to establish validity of the treatment. Adding this additional group would also increase the complexity of the study, requiring a different method of statistical analysis (ANOVA), and increase the chances of either a Type I or Type II error (Jackson, 2012).

Population

The population was adult managers from across the United States provided by Qualtrics (2014) panel. Qualtrics allows researchers to use their own population and send out the survey, it allows researchers to use Qualtrics's population and it launches the survey, or both populations with Qualtrics launching the survey and researchers sending out the survey. Qualtrics's population was from volunteers recruited from their website and their associated survey organizations. For this study, this researcher used Qualtrics to randomly select and assign from its population and launch the survey to their audience based upon this researchers target specifications. As this researcher's focus was associated with industrial and organizational psychology, the influence of managerial decision making was the primary focus, similar to what other researchers had done (Cerni et al., 2012; Curtis & Lee, 2013; Ralf et al., 2009; Seiler et al., 2010). This experimenter used Qualtrics to randomly select a representative sample from an adult managerial population and randomly assign to either a control and intervention group. The survey captured participants' demographic information so this researcher analyzed gender, education levels, and age group differences between control and intervention groups.

Sample

This researcher used Qualtrics (2014) to randomly select a sample of at least 102 participants which randomly assigned them to either a control or intervention group with each group consisting of at least 51 participants, see Figure 1 for sample size recommendations from G*Power (Buchner, Erdfelder, Faul, & Lang, 2010) calculations. Even though all panel members of Qualtrics were adults, the informed consent form required participants to acknowledge that they were adults. The informed consent form was the first page of the survey with piping logic to either the rest of the survey upon acceptance or to the thank you page upon rejection of the form. Only Qualtrics made contact and only consenting participants received a link to the questionnaire. Qualtrics recruited its panel members through partner networks. Qualtrics partner networks were an association of other providers with access to other sources of recruitment. When Qualtrics recruited its panel, panel members submitted a detailed profile of their demographics. From this detailed profile, Qualtrics filtered a population based upon this researcher's target population criteria. From this target population, Qualtrics randomly selected a sample and then randomly assigned to groups keeping the number in each group even. Qualtrics did provide eight more participants in the control group of which were included in the analysis. Qualtrics rewarded their members with e-points which could be redeemed for gift cards, airline miles, or contributions to charities.

The questionnaire asked all volunteers again to verify that they were over 18 years of age and consent to testing. All responses to answers were kept confidential by this researcher and only final summary results are published. Only Qualtrics and not the researcher has panel member identifying information. This researcher did not receive any

identifying information from Qualtrics. There was no identifying information kept within this researcher's database. All volunteers of this study's may view results along with complete instructions for discerning intuitive and analytical decision tasks on a provided link. This ensured that all participants benefit from this study regardless if they were part of the control or intervention group or completed the survey. Qualtrics recorded all answers and provided files for this researcher which he used to complete the analysis of data.

This researcher randomly selected a sample of 102 adults, who held managerial positions, through Qualtrics (2014) using their membership population. There was no requirement for length of time as a manager but they must have been in a managerial position at the time of testing. From this sample, Qualtrics performed the random assignment of participants to either a control or intervention group which reduced the nonequivalent control group threat, history effect, maturation effect, testing effect, and instrumentation effect (Jackson, 2012). This researcher excluded two responses because of four or more incomplete answers but used all of the rest of the responses in the analysis and not just the extreme scores. This reduced the regression to the mean threat. This experimenter and subjects had no communication. Only Qualtrics conducted the testing which reduced the experimenter and subjects' effect. Participants tested in a relatively short time period of a day with no repeat testing which reduced the mortality, attrition, and diffusion of treatment threat. This experimenter used measure instruments from other successful research studies which reduced the floor and ceiling effects.

Because this study used an adult limited sampling from Qualtrics (2014) members, participants were expected to be adults as they were required to be by limits set

during Qualtrics audience setup. This survey requested verification that participants were adults through the informed consent form which was the first page of the survey.

Participants who did not confirm they were adults or did not give their consent were transferred to the concluding thank you page and were not included in the intervention or survey. Through Qualtrics, this researcher randomly selected and assigned equally to either the intervention group or the control group addressing the nonequivalent control group internal validity threat. Qualtrics provided the capability of random selection and random assignment. The only issue not addressed by using this method through Qualtrics was that there was no visual contact and no photo identification used by this researcher to verify a person's age or identity.

To protect identity, only Qualtrics (2014) knew who their members were and no identifying information was returned to this researcher. No identifying information was captured in the database that this researcher used for analysis. Members of Qualtrics had a unique identifier to ensure that participants only take tests once. This researcher was not in direct visual or audio contact with participants. This study used comparisons across groups including a control group ensuring statistical equivalence through random selection and assignment. Time events and maturation threats to validity were minimal due to participants expecting to complete the whole process within an hour and the availability of the on-line system were open for a short period of time of a day. Random assignment occurred when the participant volunteered to participate which Qualtrics automatically controlled through their web site. This researcher did not know who was assigned to which group.

Figure 1**G*Power 3.1.7 Results for Two Groups Two Measures**

Input:	Tail(s)	= One
	Effect size d	= 0.5
	α err prob	= 0.05
	Power (1- β err prob)	= 0.8
	Allocation ratio N2/N1	= 1
Output:	Noncentrality parameter δ	= 2.5248762
	Critical t	= 1.6602343
	Df	= 100
	Sample size group 1	= 51
	Sample size group 2	= 51
	Total sample size	= 102
	Actual power	= 0.8058986
t tests - Means: Difference between two independent means (two groups)		
Analysis: A priori: Compute required sample size		

Materials/Instruments

This researcher tested influences of instructions for discerning between analytical and intuitive decision tasks through the survey questionnaire. Testing consisted of 10 word problem decision making tasks which include five intuitive questions or problems from De Neys and Glumicic (2008), three analytical questions from Cognitive Reflection Test (CRT) from Frederick (2005), and two additional analytical questions from Böckenholt (2012). Prior authors had given their written consent for all tools that this researcher used in this new study. After each decision task, participants indicated if they intuitively resolved the decision task or if they had to think through the problem for resolution. They indicated this through a multiple choice list which also included the option to not answer. The survey captured data through the Qualtrics (2014) web site and was stored in their password protected database on a secured server. The simple design of

this study required only a t-test for two independent group means as there were two groups with only one independent variable and one dependent variable (Black, 1999; Jackson, 2012). The tool for analysis was IBM's SPSS.

The research method in this study addressed the history effect by conducting the intervention within the questionnaire for the intervention group. By putting the treatment together with the testing it did not allow for an extended amount of time between treatment and testing. By using a control group, the subjects were probabilistically equivalent which also reduced the history effect threat (Jackson, 2012). The survey also included collecting demographic information of its participants, this researcher used this information to evaluate what other confounding factors might be present. Because this researcher did not know who the participants were, with only a post-test, and with randomly assigned participants to the two groups, the threat of regression to the mean was reduced.

Because all instruments were tested and used successfully in other prior studies (Böckenholt, 2012; De Neys & Glumicic, 2008; Frederick, 2005) and due to the use of a control group the threat of instrumentation effect was reduced. All prior authors granted their permission for their instruments to be used within this new study. Testing only once eliminated the test-retest effect. The threat of mortality and attrition was a concern of this study as participants that volunteered may drop out by not completing the survey. This study addressed this by waiting on random assignments until participants were ready to be tested. Qualtrics (2014) randomly assigned subjects as they logged on to participate. This kept the two groups close in number until both groups had at least 51 completes. Additionally, Qualtrics randomly selected more than 102 participants to reach the

recommended numbers for this study. Because the time span of intervention and testing was so small, there was little threat of diffusion of treatment (Jackson, 2012). Qualtrics allowed for survey items to be randomly ordered which this study used. All survey items were included in Appendix B.

Operational Definition of Variables

Group Type Variable, operationalized for this study identifying the group to which the participant was randomly assigned. The level of measurement of group type was nominal and varied depending on which group Qualtrics randomly assigned the participant based upon a similar study by (Kogler & Kühberger, 2007). Values are control (0) and intervention (1). The data source was from the recording of the random assignments of the volunteer participants.

System Thinking Style Choice Variable, operationalized in a nominal variable as either system one intuitive (1) or system two analytical (2). After each decision task, participants were asked to identify if they used either intuitive thinking style (“I relied on my intuitive impressions,”) or analytical thinking style (“I had to think things through carefully”) which was based upon the REI (Pacini & Epstein, 1999). Sladek, Bond, and Phillips (2010) and Witteman et al. (2009) applied REI in their research to measure participants’ inclination to use either intuitive or analytical thinking style in their decision making. Sladek et al. (2010) had 520 participants from the health profession in their study of which 77 were post-graduation students, 105 trainee doctors, 235 consultants, 50 nurses, and 53 managers. Witteman et al. (2009) had 774 university students participate in their study of which 96% were women and 97% were less than 25 years of age.

This new study used two items from REI to identify participants’ system thinking

style choice. After each decision task, there was a question that inquires how participants made their decision using the two items from REI as the only possible answers. The survey used the following example:

How did you make your decision? ____

(1) “I relied on my intuitive impressions,”

(2) “I had to think things through carefully”

Decision Type Variable, operationalized in a nominal variable of either system one intuitive (1) or system two analytical (2). There were five intuitive decision tasks as defined by De Neys and Glumicic (2008). There were five analytical decision tasks with three defined by Frederick (2005), and two additional analytical questions from Böckenholt (2012). Decision Type Variables of either intuitive or analytical have been predefined and predetermined by these other studies (Böckenholt, 2012; De Neys & Glumicic, 2008; Frederick, 2005).

Operational Definition of Constructs

Instructions for Discernment Construct, operationalized as the group type of either intervention or control as used by Kogler and Kühberger (2007). The intervention group of participants received the instructions for discerning between intuitive and analytical decision tasks in the questionnaire as used by Kogler and Kühberger (2007) and specified by (Kahneman, 2003). The control group received the questionnaire that did not have this set of specific instructions. Both groups received general instructions on how to complete the survey questionnaire.

Decision Task Type Construct, operationalized through a set of tasks which require either intuitive or analytical thinking style for resolution taken from De Neys and

Glumicic (2008) and from Böckenholt (2012). This survey used 10 decision making tasks which consist of five intuitive questions from De Neys and Glumicic (2008), three analytical questions from Cognitive Reflection Test (CRT) from Frederick (2005), and two additional analytical questions from Böckenholt (2012).

Measurement

Decision Tasks. Decision tasks were problems which require participants to use either analytical or intuitive thinking styles for resolution. Decision tasks were split into intuitive or analytical type problems. There were five system one intuitive type decision tasks based upon system one problems from De Neys and Glumicic (2008):

1. Erin is 13 years old. Erin's favorite subject is art. Erin's favorite things to do are shopping and having sleepovers with friends to gossip about other kids at school. What is most likely? a. Erin is a girl, b. Erin is a boy
2. Jay is a 29-year-old male. He has served a short time in prison. He has been living on his own for 2 years now. He has an older car and listens to punk music. What is most likely? a. Jay has a tattoo, b. Jay has no tattoo
3. Lilly is 37 years old. She is married and has 3 kids. Her husband is a veterinarian. She is committed to her family and always watches the daily cartoon shows with her kids. What is most likely? a. Lilly is a kindergarten teacher, b. Lilly is an executive manager
4. Tara is 15. She loves to go shopping at the mall and to talk with her friends about their crushes at school. What is most likely? a. Tara is a Bruce Springsteen fan, b. Tara is a Britney Spears fan
5. Martine is 26 years old. She is bilingual and reads a lot in her spare time. She is a very fashionable dresser and a great cook. What is most likely? a. Martine is American, b. Martine is French

There were five system two analytical type decision tasks of which three were based upon system two problems from Frederick (2005):

1. A bat and a ball cost \$1.10 in total. The bat costs \$1.00 more than the ball. How much does the ball cost? ____ cents
2. If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets? _____ minutes

3. In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover half of the lake? _____ days

The other two system two analytical type decision tasks were based upon system two problems from (Böckenholt, 2012):

1. Jack is looking at Anne but Anne is looking at George. Jack is married but George is not. Is a married person looking at an unmarried person? (A) Yes (B) No (C) Cannot be determined
2. A bicyclist completes a ten-mile race track with a slow average speed of 10 miles/hour. How fast should the bicyclist cycle on the second round to reach a more impressive average speed of 20 miles/hour on both rounds? _____ miles/hour speed

Choice of Thinking Style Measure. Choice of thinking style was operationalized in a ratio variable counting the number of times a participant correctly responded with choosing analytical thinking style when resolving an analytical decision task or choosing intuitive thinking style when resolving an intuitive decision task. Each decision task was either an intuitive or analytical type of decision. Participants were expected to identify if they used system one intuitive thinking style (“I relied on my intuitive impressions,”) or system two analytical thinking style (“I have to think things through carefully”) after answering each decision tasks which was based upon the Rational Experiential Inventory (Pacini & Epstein, 1999). Zero was the value captured for an incorrect system type choice for a decision task. Participants earned a score of 10% for selecting a correct thinking style. There were five system one intuitive decision tasks that required a quick intuitive system one type answer and five system two analytical decision tasks that required an analytical thought provoking system two type answer. With there being only 10 decision tasks, when participants missed one answer they earned a score of 90% correct. Percentage correct answers ranged from 0% to 100%. The data source was from the

questionnaire after each decision task question. This study produced a combined ratio score for each participant based upon the type of system assigned to the question.

Instructions for Discernment in Decision Type. Instructions for determining decision type involve four guideline instructions (intuitive versus analytical) as defined by Kahneman (2003). This new study incorporated these instructions into the intervention group's survey instructions as the intervention treatment and left off the control group's survey instructions. The first was to look for cues in the decision situation such as statistics or other facts that indicated the need for slow analytical thought. If there were no statistics or relevant facts, this was most likely a quick system one intuitive decision (Kahneman, 2011; Kogler & Kühberger, 2007). The second was to consider vigilance and accuracy. How many possible correct answers were there? If there was more than one possible answer and accuracy was not at issue, this was most likely a quick decision; otherwise, it was a slow analytical thought provoking decision. The third instruction was to the amount of time needed for the response. If the decision task could be answered easily and quickly, it was most likely a quick intuitive decision, otherwise, it might be an analytical thought provoking question. The fourth was to consider if taking a second look would help by reconsidering the response. Would a second look make a difference? Did the decision need to be rechecked for accuracy? If not, it was most likely a quick intuitive decision; otherwise, it was a slow analytical decision. The questionnaire of this study provided these instruction guidelines to the intervention group participant as an intervention before testing.

Data Collection, Processing, and Analysis

Data collection steps.

The steps for this experiment were (a) informed participants of the study, (b) secured their consent, (c) provided standard instructions on how to fill out the survey, (d) provided the intervention group special discernment instructions, (e) had participants take the survey, and (f) thanked the participants for taking the survey. The first page of the survey explained to the participants the purpose of the study and stated that they would be asked 10 multiple choice word problems to solve. The second page informed the intervention participants that after each word problem, they were to select whether they relied on their intuition or if they had thought things through carefully. On this same page, the instructions explained that participants may quit and exit the test at any time without any consequences to them. The first page included the informed consent instructions and gave the participants an option to be part of this study.

Data Collected.

The data collected included which group participants were assigned and how participants responded to how they resolved each problem. As Qualtrics (2014) members consent to testing, Qualtrics assigned them to a group type variable, operationalized for this study identifying the group to which the participant was randomly assigned. The level of measurement of group type was nominal and varied depending on which group Qualtrics assigned the participant based upon a similar study by (Kogler & Kühberger, 2007). Values were control (0) and intervention (1). The data source was from the recorded random assignments of the volunteer participants.

As for participants' responses, there were five word problems that required

intuitive thinking (De Neys & Glumicic, 2008) and five word problems that required analytical thinking (Böckenholt, 2012; Frederick, 2005). Each word problem had a decision type variable, operationalized in a nominal variable of either system one intuitive (1) for intuitive word problems or system two analytical (2) for analytical word problems. This study used decision type variables which were predefined and predetermined from other similar studies (Böckenholt, 2012; De Neys & Glumicic, 2008; Frederick, 2005).

After participants answered a word problem, they indicated how they resolved it by selecting their choice of thinking style they used. Choice of thinking style was operationalized in a ratio variable which counted the number of times a participant correctly responded with choosing analytical thinking style when resolving an analytical decision task or choosing intuitive thinking style when resolving an intuitive decision task. Each word problem decision task was either an intuitive or analytical type of decision.

Participants identified if they used system one intuitive thinking style (“I relied on my intuitive impressions,”) or system two analytical thinking style (“I have to think things through carefully”) after answering each decision task which was based upon the Rational Experiential Inventory (Pacini & Epstein, 1999). Incorrect answers resulted in a score of zero for a decision task. When participants did not mark decision tasks with correct responses, their score was a zero value. Skipped word problems or questions resulted with the same score as not marking responses and participants earned a zero value. When the choice of thinking style was correct participants scored 10%. There are five system one intuitive decision tasks that required a quick intuitive system one type

answer and five system two analytical decision tasks that required an analytical thought provoking system two type answer. The test captured the participants' percentage of correct answers which ranged from 0% to 100%. The data source was from the questionnaire after each decision task question during testing. This researcher evaluated this against the type of system assigned to the question and produced a combined ratio score for each participant.

This researcher did not use survey results when participants exited the survey without completing it. Qualtrics (2014) continued to supply participants until the required number of completed surveys was achieved. The simple design of this study required only a t-test for two independent group means as there were two groups as the one independent variable and one system choice type as the dependent variable (Black, 1999; Jackson, 2012). The tool for analysis was IBM's SPSS.

Assumptions

The sampling of participants were managers which impacts generalization of this study to other professions and non-adult populations. Because managers were involved in logical analytical work from time to time, they might have been more inclined to incorporate system two analytical thinking and more susceptible to instructions for discerning between analytical and intuitive decision tasks. This study addressed this threat by having two groups of which one was a control group to establish a baseline for comparison with the treatment intervention group (Black, 1999).

This researcher assumed that intervention participants read the discerning instructions before continuing with the testing. When participants did not read or apply the instructions they invalidated their responses. Intervention participants had to

acknowledge that they read and understood the instructions before they took the test. When they did not wish to acknowledge that they read and understood the instructions they exited the testing and transferred to the thank you page.

Qualtrics (2014) assumed their audience was truthful and did not offer large cash rewards to participants. Their contribution to their audience was minimal. To ensure data quality, Qualtrics screened their audience, used profiling for targeting, verified using social networking, and inferred information through other websites. Some of the major concerns Qualtrics strived to eliminate were inaccurate information, duplicate accounts, and invalid responses. Additional concerns were participants hurriedly completing surveys, not considering all options, choosing the same option for all questions, and personal biases. Qualtrics contended that the incentive process minimized data quality issues and encouraged honest responses. Qualtrics provided this information to this researcher.

Limitations

Because the population participating in this research included only those in management, there is a limitation of generalization to others who have people who directly report to them and are decision makers. Other researchers would need to investigate samples from other populations before generalizing beyond management. Because this researcher's area of focus was in industrial and organizational psychology, this researcher purposefully limited the study to those who managed and made decisions for their organizations.

This study used a post-test only control group design which was a between-subjects design and reduced testing effects which was a significant strength (Jackson,

2012). The weakness of this design was that more participants were needed, it took more time to execute, and it reduced the statistical power. The advantage of this design was that instructions for determining between both intuitive and analytical decision tasks were tested for both types of decision tasks. The control group provided a group to compare with the intervention group for significant difference. Providing this treatment for both intuitive and analytical decision tasks was a new approach that other studies had neglected to incorporate in their design. This provided an understanding to the effects of instructions for discerning between analytical and intuitive decision tasks as treatment upon participants' choice of thinking style. Providing system type identification for only system two analytical problems was not new as it was performed in another study by Kogler and Kühberger (2007). The prediction for this study was that these instructions in discernment influenced participants' choice appropriately using system one thinking style for intuitive decision tasks and system two thinking for analytical tasks.

Delimitations

Dual process and system theory covers many areas of interest. As this researcher was associated with industrial and organizational psychology, the influence of managerial decision making was the primary focus as other researchers have shown (Cerni et al., 2012; Curtis & Lee, 2013; Ralf et al., 2009; Seiler et al., 2010). Prior theories confirmed the need for controlling intuitive belief-biases (Andrews, 2010) and analytical biases (Goldsmith & Cyboran, 2013). Due to these biases, people face a dilemma they override the appropriate choice of system thinking for a decision task. There has been some success in the past with influencing these biases (De Neys & Glumicic, 2008; Kogler & Kühberger, 2007). Results supported that identifying analytical decision tasks improves

decision accuracy. What researchers did not investigate was the influences of system one intuitive type instructions on intuitive type decisions. This research incorporated intuitive and analytical instructions for resolution of both intuitive and analytical decision tasks which provided a more holistic approach in overcoming both intuitive and analytical biases. Because there was already support for the effects of system type identification upon decision accuracy (Andrews, 2010; Goldsmith & Cyboran, 2013; Kahneman, 2011; Kogler & Kühberger, 2007; Stupple et al., 2011), this research focused on understanding the participants' choice of system thinking they used for problem resolution when provided with discernment instructions.

Ethical Assurances

Compliance standards.

Ethical behavior promotes credibility by scientists being open and honest about their sources, research, methods, procedures, findings, and conduct to respectfully add to the overall body of knowledge (Committee on Science et al., 2009). This author worked to ensure that the conception, planning, design, execution, and reporting of his doctoral research was ethical, honorable, credible, valuable, and respectable by taking the following actions.

Plagiarism

Ethical researchers used only scholarly peer reviewed articles from reliable sources (Gruber, Knefel, & Waelchli, 2008). This required going beyond marking the check boxes for scholarly and peer reviewed articles on the on-line databases. It required that this researcher evaluated the articles looking to see if articles went through peer review; understanding the articles' audience; checking if biases were present; and

investigating if the work provided a way to validate their claims.

This author included appropriate related research in his literature review which include positions supporting or contrary to his contentions. This author did not violate copy right laws by making copies or disbursing any articles with others. He only shared his sources for others to contact and request their own copies for their verification of his research and for their future research.

This author appropriately cited all articles in American Psychological Association (2010) format even if the work was his own, in order to prevent plagiarism (Committee on Science et al., 2009). He either encased all quotations in quotation marks or indented paragraphs to ensure readers know when someone else's work is being presented. He paraphrased the works of others in his own words, cited sources appropriately, and others verified his writings through on-line plagiarism checker tools such as Turnitin (iParadigms, 2013). He corrected all known discrepancies before any peer reviews were conducted. He made corrections of known errors before any materials were provided for review. Once published, if anyone identifies discrepancies, he will take whatever ethical and legal actions are necessary to correct the situation which may include retraction.

Risk assessment

Innovations as well as any research has risks which effective researchers prepare for in advance of their study (Committee on Science et al., 2009). Researchers must consider risk of harm to human or animal subjects which may be physical or mental (American Psychological Association, 2013). Even though there may be no physical or mental harm, there may be perceptions of discomfort which may discourage participation in future research. This experimenter did not put people in risk of losing their job,

harming their reputation, suffering from criminal actions, civil penalties, or financial loss due to participation.

This study's research team did conduct an initial risk assessment and had the proposal reviewed by the institutional review board (IRB). This researcher identified all foreseeable risks, made an assessment as to probability and impact of each risk, and developed a mitigation strategy for those risks with high or medium probability or impact. The team reviewed the low probability and impact of risks on a weekly basis which ensured that no change has occurred in their assessment.

The risk assessment verified that participants were not vulnerable to harm, risks were minimal, and protections were in place. This mitigation strategy was in place before experimentation began as specified by Yanos, Stanley, and Greene (2009). The following were risk mitigation strategies for this doctoral study which address informed consent, data collection, privacy, confidentiality, conflicts of interest, data handling, reporting, mistakes, negligence, mentorship, review board, editors, and consultants.

Informed consent

Informed consent is a process (Plankey-Videla, 2012) that helps participants understand how they will be involved, what intervention will be performed on them, the research limitations, and any risks they might incur (Nunan & Yeniciglu, 2013). This author based his informed consent form on the Northcentral University (2013d) sample informed consent form. This form included information about the study concerning system thinking; how people participate in taking a test; what interventions they received concerning system thinking instructions; that there was no risk of harm to them; and they were under no obligation to participate or complete the testing even after they begin.

The population for this study was from Qualtrics (2014) members. It was up to the president of the organization if he or she wants the name of the organization included in the report as specified by (Northcentral University, 2013a). Qualtrics has already provided a permission which specified their expectations and agreements. This researcher provided the informed consent form as the first page of the on-line survey.

Those members who wished to participate acknowledged that they understood the purpose and expectations of this study and electronically signed the consent form which captured their consent within the database. Members acknowledged that they understood that they may opt out at any time without any negative consequences in order to participate and that their participation was purely voluntary. Even though there were exceptions for not using informed consent, this study does not meet any of those exceptions because it was not a naturalistic observation study nor did it require any type of deception (American Psychological Association, 2013).

The informed consent form is available in Appendix B. The first page of the survey was the informed consent page which explained the name, purpose, and process of the experiment in common language. The name of this study was “Effects of Instructions for Discerning Analytical and Intuitive Tasks”. The purpose of the study was to understand how people think when presented with instructions. This survey was an invitation for people to participate in a research study as part of the dissertation process at Northcentral University in Prescott, Arizona. Qualtrics (2014) selected adult managerial participants from their voluntary membership as specified by this researcher.

The survey presented 10 word problem questions and asked for participants for their resolution by selecting one from a list of multiple answers. For every word problem,

participants were able to select that they did not wish to answer. There were no time limits for each problem. This researcher estimated the survey took on average about 15 minutes to complete based upon personal practice and the number of questions. Other researchers used these word problems in their research before with no threat to participants. The survey explained to participants that they were expected to solve each problem on their own without help from others. After each word problem, participants chose an option describing how they went about solving the problem or they marked they did not wish to answer. At any time, participants were able to quit the testing without any consequence to them. There were no known risks in participating in this research study and there were no questions of a personal nature asked. Participants were able stop the survey at any time or simply not answer a question that made them feel uncomfortable in answering.

The benefit of this study was to help this researcher understand the effectiveness of instructions in resolving word problems. Participants may have directly benefited from this study as instructions were designed and have been used to assist in decision making. This research did not require or use any personal identifying information. All data collected was kept confidential and no personal identifying information was captured in the survey. Summary results were made available to all participants once the survey concluded.

Being a part of this survey was voluntary. Compensation was only through Qualtrics and its method of contributing to participants' charities and its weekly sweepstakes process. When participants chose to continue with the testing, they pressed the accept button which confirmed they had read all the information, were voluntary

participants, and were at least 18 years of age. When participants did not wish to continue with the testing, they pressed the cancel button without any consequences.

Those who did not wish to give their consent were transferred to the thank you page and the survey ended. For those participants who gave their consent and were part of the intervention group, the next page of the survey was the four set of instructions for discerning between intuitive and analytical decision tasks. For those who gave their consent and were part of the control group, they were transferred to the third page, bypassing the special intervention instructions on the second page, and began taking the survey. Once participants completed the survey, the thank you page appeared and the survey ended. Once all participants completed the survey, Qualtrics (2014) notified this researcher to download results for further analysis within SPSS.

Data Collection

Northcentral University Institutional Review Board (IRB) reviewed the research and gave its approval before this researcher began collecting any data (Northcentral University, 2013c). This researcher conducted this experiment through Qualtrics (2014) and all participants acknowledged that they were adults. This researcher had Qualtrics randomly assign participants to either a control or intervention group.

For the control group, Qualtrics (2014) bypassed the discernment instructions while the intervention group received discernment instructions as the intervention, along with the test. Qualtrics compensated the participants with points which the participant could use for rewards or contribution to charity for their time in answering the survey. All participants who consented to the testing whether they completed it or not received the compensation. Other studies used all testing and survey items within this study without

any harm to participants which mitigated the risk of physical and mental harm.

Privacy and confidentiality

Privacy in research involves guarding personal information by preventing unauthorized access and only providing the information that others need to know (Francis, 2008). Confidentiality is the sharing of information with only those who are authorized. In the case of this doctoral study, this researcher did not need personal identifying information, academic records, financial statements, or any health information. Northcentral University (2013a) explained that confidentiality involves researchers protecting participants' identity from outsiders. Using pseudonyms or sequence numbers helps protect those from the outside knowing the identities of participants. This study hid the names of participants and only referred to them by a sequential number which Qualtrics assigned.

For this study, only Qualtrics (2014) made direct contact with and knew the identity of the participants which protected their anonymity. Qualtrics scrubbed all identifying information, like email addresses and names, and ensured there were no duplicate participation by subjects by checking the IP addresses for duplicates. Only Qualtrics kept the assigned identification number cross reference. This researcher did not have direct contact with participants.

Participants electronically signed their confidentiality agreement forms. Once all participants completed the survey, Qualtrics passed on the anonymous testing and survey data to this researcher. Qualtrics captured informed consent data for auditing of this study to ensure confidentiality and audit information.

This researcher placed the password encrypted file on their secure file share

system to be retained for 7 years and set to automatically delete after that time. This researcher's file share system is password protected and will automatically delete the information after the 7 years pass. This author will ensure that the deletion occurs once the 7 years has passed.

This researcher disclosed his finding without including any information that would identify the participants. He presented results in an aggregated manner without personal identifications. Once the study was concluded, this researcher made all intervention materials and summary results available to all who participated in the study through a link page on Qualtrics. Only Qualtrics (2014) made contact with the participants providing them with this information.

Conflicts of Interest

No one on the research team had any conflicts of interest which compromised their involvement or influenced personal gain (Horner & Minifie, 2011). The research team identified and enacted a strategy that addressed any potential conflicts of interest through the risk assessment process. This researcher assessed the probability of any conflict occurring along with any impact and took mitigating actions to minimize the risk. Regardless of any additional actions, all potential or actual conflicts of interest were disclosed. In case of a conflict of interest, the team would have swiftly disclosed the conflict to the mentor and IRB and dealt with the issue for resolution which might have resulted in dismissal and replacement of any team member from the study, including Qualtrics staff.

This researcher was not a member of Qualtrics (2014) while the study was in progress to avoid any conflict of interest (Committee on Science et al., 2009). Qualtrics

only facilitated the experiment and did not directly benefit from the study any more than the general public, to avoid any conflict of interest. Qualtrics was paid 1,275 dollars for recruiting a panel and to compensate the panel for their participation. The IRB monitored the study as this researcher executed his research.

Qualtrics (2014) charges \$1,275 USD for recruiting 102 panel members of managerial professionals to participate in this study. Other than the \$1,275 USD, there was no exchange of money. Qualtrics was responsible for protecting its member privacy and agreed with helping in this research study. The membership had a history of participating in surveys and research studies in the past. Members understood and expected researchers to ask for their participation knowing that they were free to decline.

Data handling and reporting

Qualtrics (2014) collected and scrubbed any identifying information and handed off the demographic and testing data to this researcher for analysis. This researcher accounted for and disclosed any missing data as specified by Sterner (2011). This was included in the reporting, how the information benefited the study, how missing information affected distribution, how many records were missing data, what actions this researcher took to manage without the data, and why actions were taken.

This researcher stored the information in a password protected secure database. This researcher used his own user identification and password for the system to audit and track his actions of retrieval, modification, aggregation, and removal of any information. As this researcher executed research steps, he published weekly progress reports to Northcentral University stakeholders through his chairperson. Once the research concluded, the author made available all results to those who participate and all

stakeholders. Additionally, so that all participants benefit from the intervention, this researcher provided the intervention instructions to the control group participants for their benefit.

Mistakes and negligence

This researcher ensured that he recorded all actions through an audit log so that others may verify and repeat all analytical actions. The author of this doctoral study accepted full responsibility for all actions of all experimenters, researchers, and participants in execution of this study and future studies as set forth by the Committee on Science et al. (2009) and the American Psychological Association (2013). This author strived to ensure accuracy through careful and methodical conduct and ethical behavior with multiple peer, mentor, and IRB reviews. This researcher gave visibility in his weekly reporting of any mistakes that occurred and the actions taken to remedy the situation. This author ensured that all actions including mistakes were published in the final report.

Working with a Mentor

This author relied heavily on his chairperson for advisement, counsel, guidance, and leadership throughout this process (Committee on Science et al., 2009). This author strived to make this relationship beneficial to his chairperson by listening and incorporating the chairperson's counsel, ideas, and direction. This author also ensured that the mentor received proper credit in the publication of the study.

The chairperson established her role and expectations at the beginning of the relationship of which this author expects to maintain throughout the study and into future research studies (Committee on Science et al., 2009). At the beginning of this

relationship, both, chairperson and this author arranged communication paths and recurring times for advisement and review of this study. The chairperson was the first to have the opportunity to review all proposals and procedures as this author recognizes that he was new to the doctoral process and can gain greatly from the knowledge and experience of his chairperson.

Northcentral University requirements for IRB approval

Northcentral University (2013b) is required by law to have an institutional review board (IRB) and review all research activities that researchers are planning on conducting on human subjects. Northcentral University's IRB approved this doctoral research study before any data collection began. This author expected the IRB to monitor the study as this experimenter conducted research and when this researcher publishes his findings. This author valued the IRB as part of the doctoral research study. This researcher was swift to incorporate any changes the IRB recommended or required as the chairman reviewed and instructed him.

Summary

Prior research neglected the need for system one intuitive type decisions and primarily focused on improving only system two analytical thinking. Instructions in identifying decision types had only been tested as it influences system two analytical type decisions which presented a bias toward analytical thinking and created a conflict when intuitive thinking was all that was required to resolve the problem. This researcher needed to test instructions which determine difference between both intuitive and analytical decision tasks rather than assuming the system two analytical type was all that was needed. The purpose of this study was to measure how instructions that distinguish

differences between analytical and intuitive decision tasks influenced a person's choice of system thinking style for resolving decision tasks.

As a scientist, this author was prepared and plans to be ethical throughout his career. His desire was to establish and maintain the trust of his colleagues and preserve his own personal honor. He strived to significantly contribute to the overall body of knowledge serving the public (Committee on Science et al., 2009). His first and foremost goal was the safety and care of the participants of this study by identifying risks, assessing the probability and impact of those risks, and preparing a mitigation response. He will continue to maintain throughout his career the mandate to do no harm.

Through ethical conduct, this author will establish credibility with other scientists. His research was open and honest about sources, methods, procedures, and findings (Committee on Science et al., 2009). From conception and throughout planning, designing, executing, and reporting, this author conducted himself ethically and honorably. He utilized the informed consent process and ensured the identity of participants remained confidential. He minimized and addressed any conflicts of interest. Additionally, he handled data appropriately being open and honest in reporting. He worked to minimize mistakes but when they occurred, he quickly identified them and worked towards a resolution that addressed them appropriately. For this author, Northcentral University's institutional review board played an important role in his research success.

Chapter 4: Findings

The purpose of this quantitative study was to gain understanding in how instructions for discerning between analytical and intuitive tasks affected participants'

choice of thinking style for their problem resolution. In order to accomplish this purpose, participants were split into two groups which were a control group and intervention group. Both groups read the standard instructions for taking the survey. The intervention group read cognitive instructions for discerning between analytical and intuitive type decisions. The control group did not have the cognitive instructions. The survey provided 10 decision tasks of which five required intuitive thinking to resolve and five required analytical thinking to resolve. Participants were to resolve the questions and then state which system thinking style they chose to resolve the question. Participants scored points for correctly selecting analytical thinking for analytical questions and intuitive thinking for intuitive questions. Participants did not earn points for skipped or for selecting the incorrect thinking for the associated decision tasks. The analysis used an independent t-test to evaluate participants' responses between the control and intervention groups. This chapter presents results, evaluates findings, and summarizes these results.

This researcher used a t-test to answer the question how instructions for discerning between intuitive and analytical decision tasks would impact a person's choice of system thinking style for resolving both analytical and intuitive decision tasks. The null hypothesis was that the intervention group's mean for their choice of system one thinking style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically equal or less than the control group's mean for their choice thinking style, $H_0: \mu_0 \leq \mu_1$. The alternate hypothesis was that the intervention group's mean for their choice of system one thinking style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically higher than the control group's mean for their choice of thinking style, $H_1: \mu_0 > \mu_1$.

The survey presented three additional questions to collect data on gender, age, and education level. This chapter presents finding on t-test results for gender. This researcher also evaluated the effects of age and education levels on this study. Ages were grouped into two groups. Using 40 as the average mid-life divider, the two groups were those who were below 40 years of age and those who were 40 or greater. In order to understand the impact of higher education, education levels were divided into two groups of those without college degree and those who completed college.

Results

The result of the independent samples t-test showed a significant difference between the control and intervention groups. On average, intervention participants scored higher ($M = 73.53$, $SE = 2.37$) than control participants ($M = 65.42$, $SE = 2.45$). The difference between the groups was significant $t(108) = -2.36$, $p \leq .05$; it represented a small-sized effect $r = .22$. The results supported the alternate hypothesis for this one tailed test as shown in Table 1 and Figure 1.

Table 1

Descriptive Statistics by Control and Intervention Groups

Group	<i>n</i>	<i>M</i>	<i>SD</i>	<i>SE</i>
Control	59	65.42	18.78	2.45
Intervention	51	73.53	16.95	2.37
Total	110			

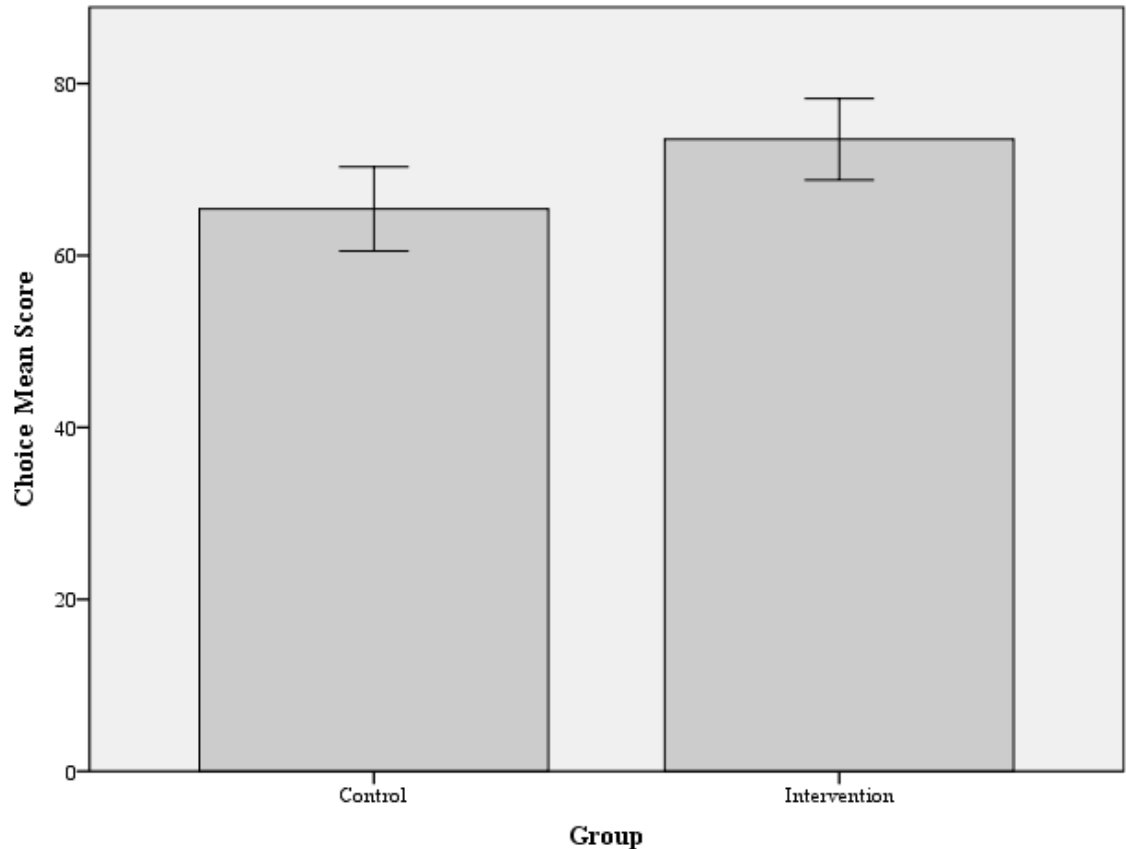


Figure 1. System thinking choice scores mean differences for control and intervention groups with error bars ± 2 SE.

To further break down the decision tasks into analytical and intuitive, a separate t-test was conducted for both decision tasks. While this was not the research question and no hypothesis was made as to the outcome, this information might lead to future research and understanding. An independent t-test analyzing the control and intervention groups for only intuitive decision tasks revealed a significant difference between the control and intervention groups for intuitive decision tasks only. On average, intervention participants scored higher on intuitive decision tasks ($M = 36.47$, $SE = 2.22$) than control participants ($M = 30.34$, $SE = 2.50$). The difference between the groups was significant $t(108) = -1.81$,

$p \leq .05$; it represented a small-sized effect $r = .17$ as shown in Figure 2.

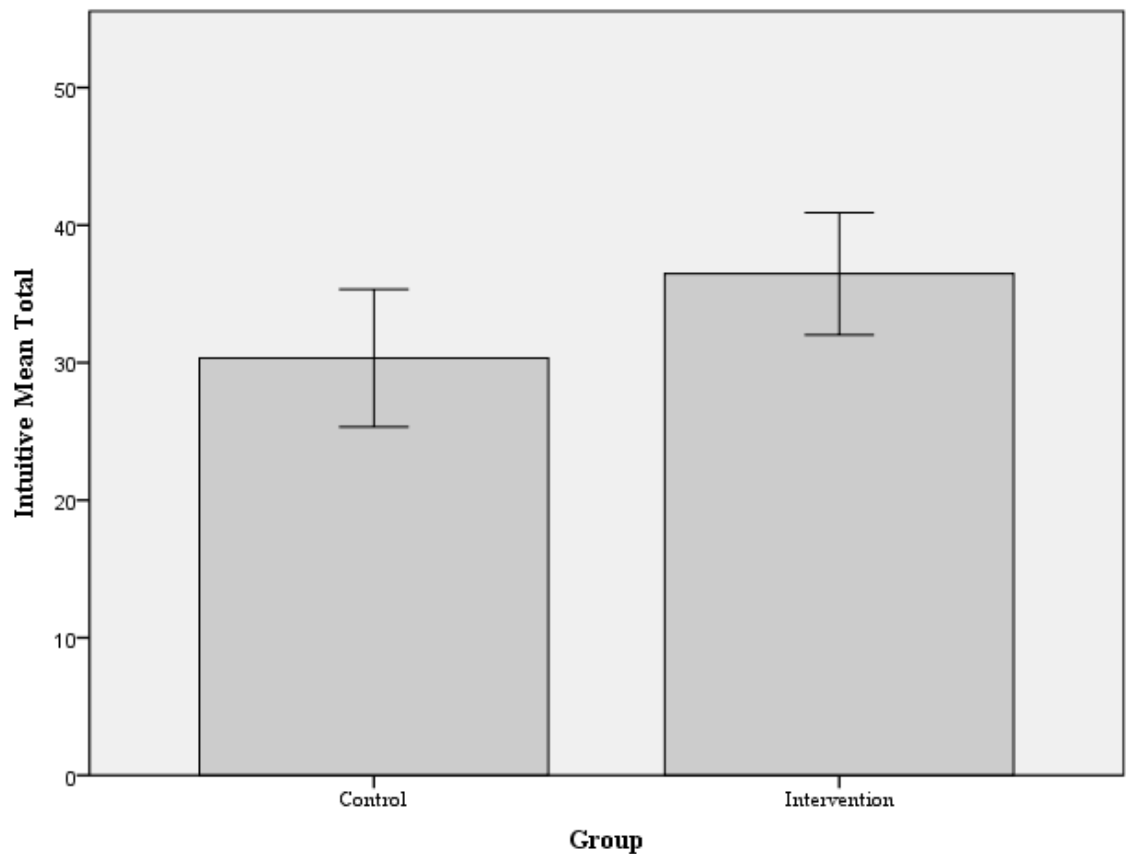


Figure 2. System thinking choice mean scores for intuitive decision tasks only by control and intervention groups with error bars ± 2 SE.

An independent t-test analyzing the control and intervention groups for only analytical decision tasks revealed a difference between the control and intervention groups for intuitive decision tasks but it was not significant. On average, intervention participants scored higher on analytical decision tasks ($M = 73.53$, $SE = 2.373$) than control participants ($M = 65.42$, $SE = 2.445$). The difference between the groups was not

significant $t(108) = -.65, p > .05$; it represented a small-sized effect $r = .06$ as shown in

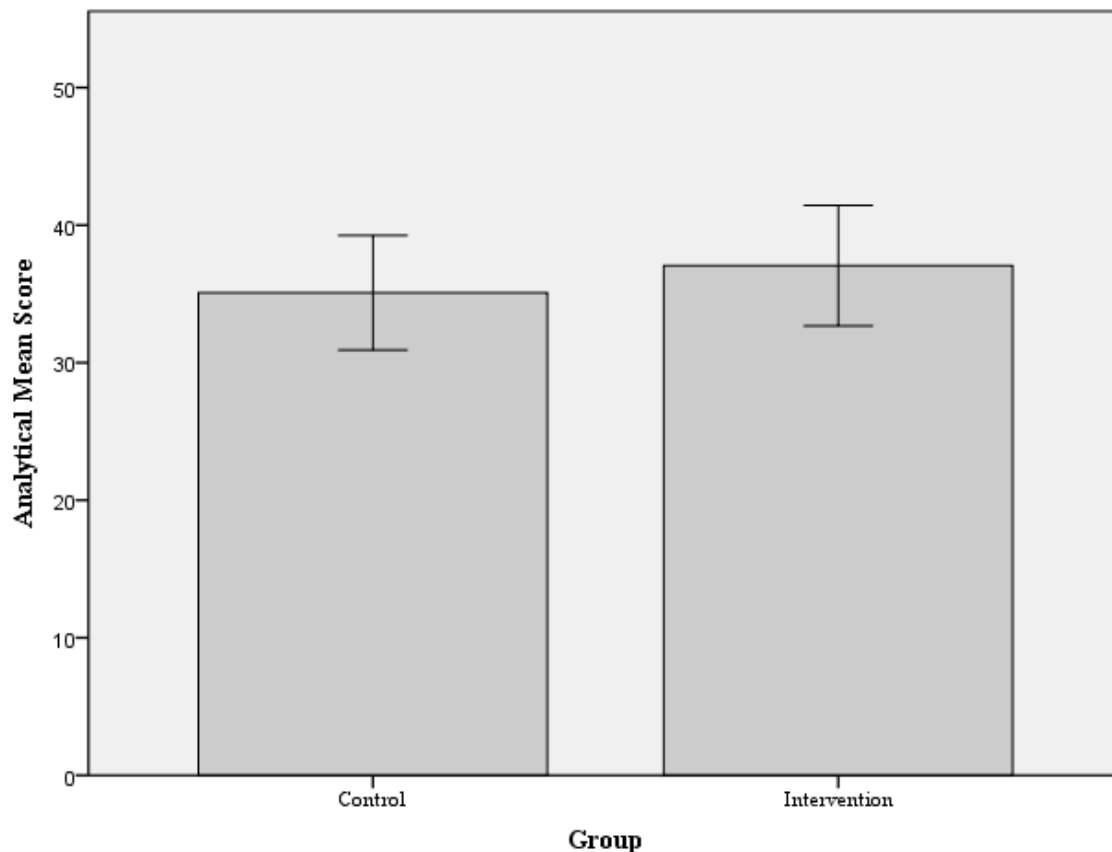


Figure 3. System thinking choice mean scores for analytical decision tasks only by control and intervention groups with error bars +/- 2 SE.

While there was not a hypothesis made about gender, age, or education level. This researcher also conducted an analysis of data for each of these areas. In the control group, one participant did not state gender. In the intervention group, all participants stated their gender. Participants in both groups stated their age. Participants in both groups stated their education level. Results showed no significant difference between female and male response within control or intervention groups as shown in Table 2 and Figure 4.

Table 2

Descriptive Statistics for Gender by Control and Intervention Group

	Control		Intervention		Total	
Gender	n	M (SD)	n	M (SD)	N	M (SD)
Male	27	64.07 (20.99)	17	70.59 (18.86)	44	66.59 (20.22)
Female	31	67.42 (16.52)	34	75.00 (16.00)	65	71.38 (16.57)
(Blank)	1				1	
Total	59	65.42 (18.78)	51	73.53 (16.95)	110	69.18 (18.33)

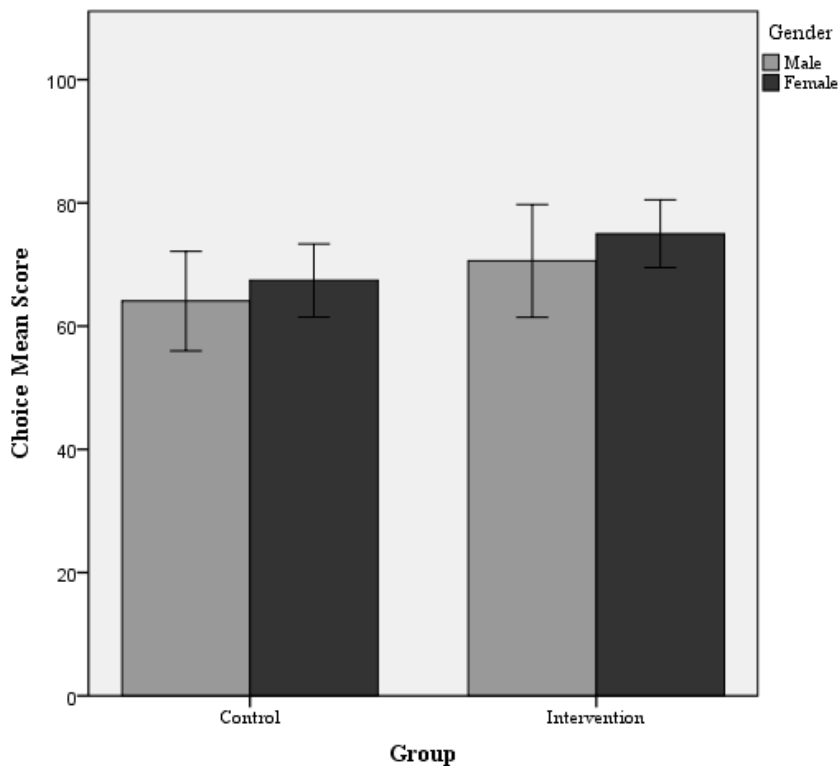


Figure 4. System thinking choice mean scores for gender by control and intervention groups with error bars ± 2 SE.

Results showed no significant difference between responses for those below 40 years of age and those above 39 years of age within the intervention group as a whole and shown in Table 3 and Figure 5. There was a difference for below 40 and above 39 years of age within the control group which may be an area of study for future research see

Figure 6. Those below 40 years of age in the control group scored higher than those in the 39 years of age while in the intervention group results were reversed which together brought no significant difference between age overall as shown in Figure 5.

Table 3

Descriptive Statistics for Age by Control and Intervention Group

Age Group	Control		Intervention		Total	
	n	M (SD)	N	M (SD)	n	M (SD)
Below 40	29	70.00 (20.35)	24	72.08 (16.93)	53	70.94 (18.73)
Above 39	30	61.00 (16.26)	27	74.81 (17.18)	57	67.54 (17.95)
Total	59	65.42 (18.78)	51	73.53 (16.95)	110	69.18 (18.33)

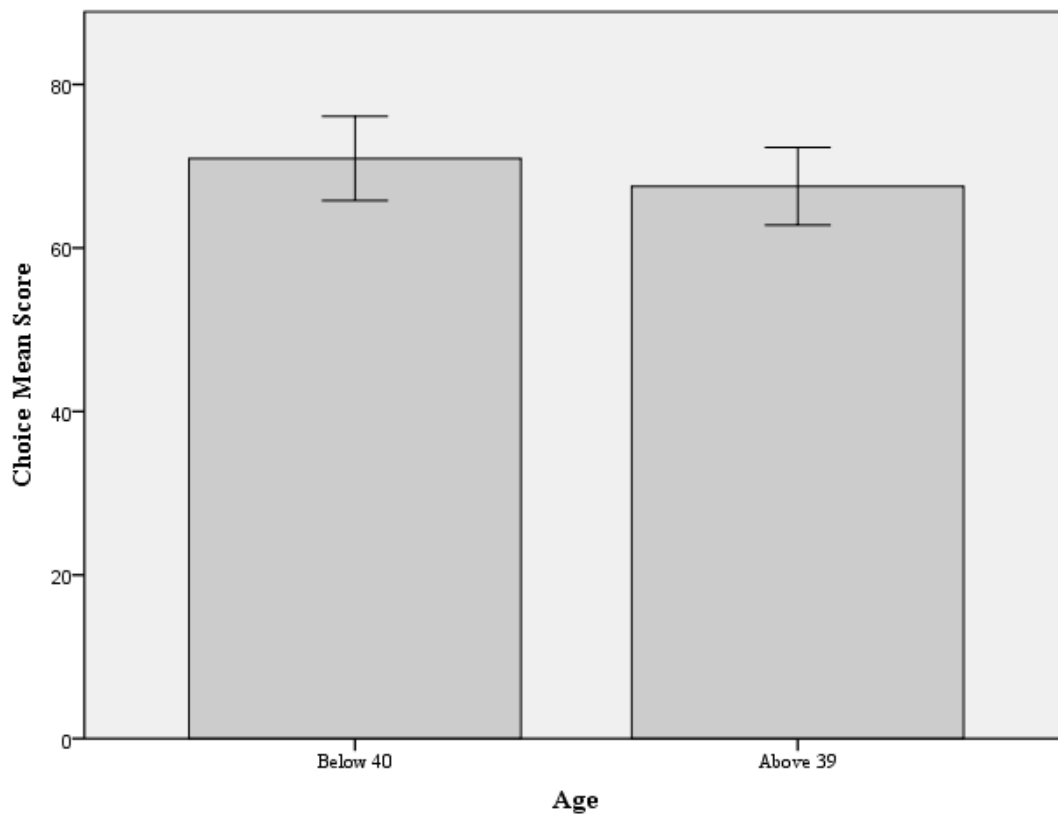


Figure 5. System thinking choice mean scores by age broken out by below 40 and above 39 with error bars +/- 2 SE.

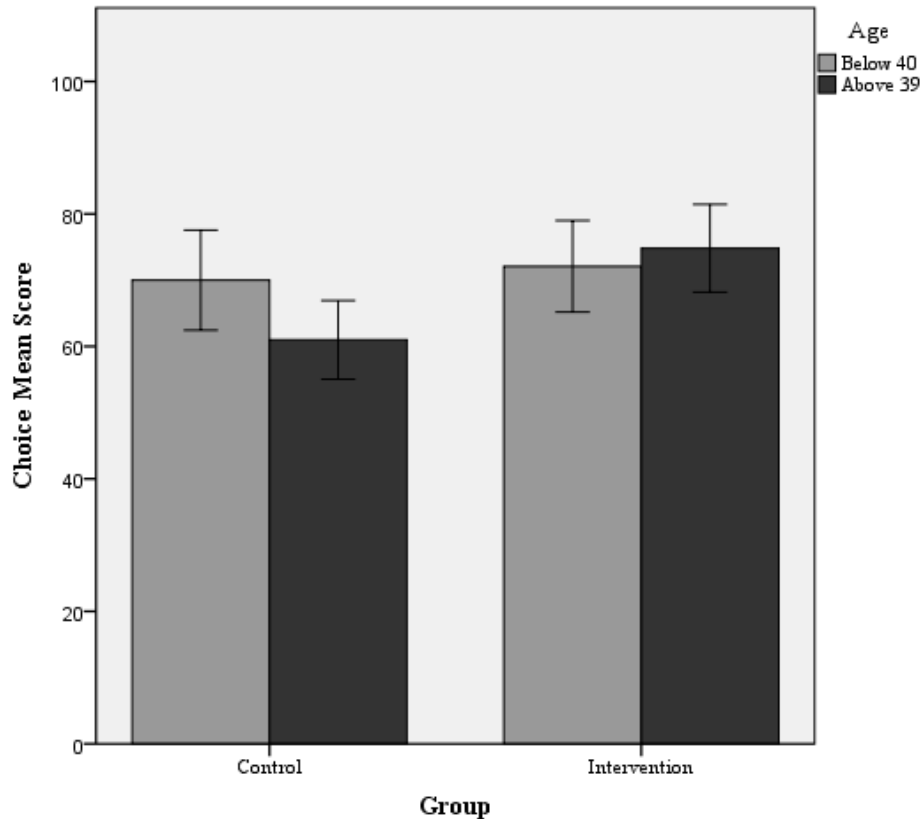


Figure 6. System thinking choice mean scores for age by control and intervention groups with error bars ± 2 SE.

Results showed no significant difference between responses for those who completed high school and those who attended college as a whole and shown in Table 4 and Figure 7. Minor differences are present but with uneven numbers between groups and education levels, results are inconclusive. This may be an area for further research. In all cases, the intervention means was greater than the control means regardless of the breakdown by gender, age, or education level.

Table 4**Education by Control and Intervention Group with Number and Mean Score**

Education	Control		Intervention		Total	
	n	M (SD)	N	M (SD)	n	M (SD)
High School	9	64.44 (18.78)	10	78.00 (17.51)	19	71.58 (18.93)
College	50	65.60 (18.97)	41	72.43 (16.85)	91	68.68 (18.27)
Total	59	65.42 (18.78)	51	73.52 (16.95)	110	69.18 (18.33)

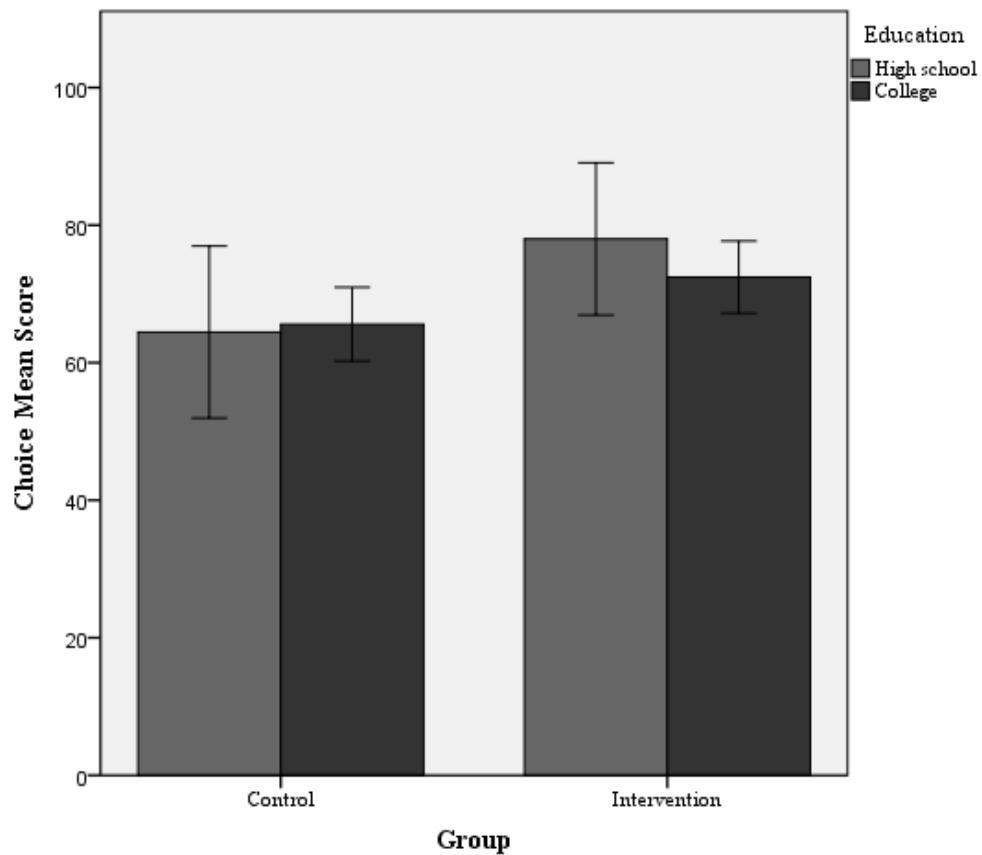


Figure 7. System thinking choice mean scores for education level of high school or college by control and intervention groups with error bars ± 2 SE.

In all cases, Levene's test was not significant which assumed homogeneity of variances was met which required the test statistics from equal variances assumed from SPSS. As the hypothesis was a one-tailed test, the Sig (two-tailed) was divided by two to determine p significance. In the case of this study's hypothesis, significance was met without dividing by two.

Evaluation of Findings

People think in basically two different ways: analytical and intuitive (Kahneman, 2011; Witteman et al., 2009). Sometimes they think quickly when they should take more time and sometimes they think slowly when it is not needed. The expectation of this researcher was that intervention participants would use the cognitive instructions to help them choose the correct system thinking style for the associated decision task. The results supported that intervention participants' mean scores were higher than control participants for choosing the correct system thinking style for the associated decision task type.

The further breakdown of the results by gender, age, and education did not show any significance between females and males; those 40 and over versus those under 40; those who had a college education and those who did not. What was unexpected was that intervention had more of an effect on participants' choice of system thinking for the intuitive decision tasks than the analytical decision tasks.

Researchers had provided training and instruction as interventions in prior studies (Klossek et al., 2011; Schoenmakers et al., 2010). This new study built upon prior research and incorporated instruments used previously to bring about results (Böckenholt, 2012; De Neys & Glumicic, 2008; Frederick, 2005). The intervention instructions for

identification of analytical decision tasks were used by Kogler and Kühberger (2007) and specified by Kahneman (2003). Results confirmed the expectations as those participants who had the intervention cognitive instructions scored higher on choice of system thinking style. As people have to make a conscious and deliberate effort to develop and use system two analytical thinking style (Kahneman, 2011), what happens when both analytical and intuitive decision tasks are randomly presented? What happens when discernment instructions explain what to do for both analytical and intuitive decision tasks? The purpose of this study was to investigate these questions and answer how instructions for discerning between intuitive and analytical decision tasks influenced a person's choice of system thinking style for resolving analytical and intuitive decision tasks. Additionally, this study focused on participants' choice of system thinking style while prior research focused on decision accuracy. By using choice of system thinking style, participants did not have to have strong math or logic skills as only their selection of thinking style was evaluated.

Summary

The purpose of this quantitative study was to gain understanding in how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for their problem resolution. Results supported the alternate hypothesis that the intervention group's mean for their choice of system one thinking style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically higher than the control group's mean for their choice of thinking style. Expectations were supported by other studies because all instruments and intervention instructions were previously used in other research (Böckenholt, 2012; De

Neys & Glumicic, 2008; Frederick, 2005; Kahneman, 2003; Kogler & Kühberger, 2007).

The result of the independent samples t-test showed a significant difference between the control and intervention groups' mean scores for correct choice of system thinking style.

Chapter 5: Implications, Recommendations, and Conclusions

There are few studies which addressed how training and instruction improves choice of thinking style (Barton et al., 2009; De Neys & Glumicic, 2008; Stuppel et al., 2013) and most of these studies only address the analytical side. The purpose of this new study was to address the problem of understanding how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for problem resolution. As the population examined included only those in management, there was a limitation of generalization to only adults who have people directly reporting to them. This chapter provides implications of the results, recommendations for future research, and conclusions drawn from this research.

Implications

The implications for industrial and organizational psychology centers on improving corporate decision making. Organization leaders frequently face business decisions of which require quick or in-depth thinking. Decisions which are preference based, creative, subjective, or conditioned only need intuitive system thinking. Decisions which are evidence based, analytical, objective, or calculated need analytical system thinking. This research provides the leader with cognitive discerning instructions which they may use to guide them in determining if they should quickly answer or take time to think through the evidence.

Prior research used instructions as an intervention to influence analytical thinking but did not include intuitive thinking. Schoenmakers et al. (2010) hypothesized that attention bias reduction training would reduce the influence of alcohol cues. This research focused on the analytical side of controlling the influence of alcohol cues and

addiction for heavy drinkers. De Neys and Glumicic (2008) instructed students to read problems aloud and expected students would improve their ability to answer analytical problems correctly. Kogler and Kühberger (2007) experimented with instructions for identifying analytical type decision tasks and discovered that people improved their accuracy when they recognized the decision task required analytical thinking. These studies did not focus on improving the use of intuitive thinking while this new study included both intuitive and analytical thinking. Prior researchers assumed because the thinking style was intuitive, no instruction was needed. What prior research missed was what happens when intuitive questions are mixed in with analytical questions. Would participants perform as well? The results of this new study support the use of instructions for influencing a person's choice of system thinking style, especially with intuitive decision tasks. These prior studies evaluated the effectiveness of the intervention based upon the correct number of answers participants provided. They did not evaluate if participants used the analytical thinking to respond to the decision tasks. This study was able to show the influences of the cognitive instructions upon choice of system thinking style rather than decision accuracy.

Prior research focused on decision accuracy rather than choice of thinking style. Kogler and Kühberger (2007) showed that when people used their intuitive thinking for analytical decision tasks, their accuracy decreased due to their biases. Cools and Van Den Broeck (2008) showed that those who used analytical thinking for intuitive decision decreased accuracy due to prolonged or overthought responses. These studies measured correct answers, but not participant's choice to think analytically. Decision accuracy was problematic in that participants must have had skills to resolve tasks correctly (Thompson

& Johnson, 2014). Some chose the correct style of thinking but still answered the question incorrectly due to their knowledge and understanding of the problem. In this new study, decision accuracy was not measured while choice of system thinking was. By measuring choice of thinking style, participants did not need to be skilled in the decision tasks areas. According to the results, intervention participants scored higher than control participants on choice of thinking style especially on intuitive decision tasks which did not require skill to answer. Prior research missed this discovery because of focusing on decision accuracy and analytical thinking style only. By including both analytical and intuitive thinking in both the instructions and decision tasks and by focusing on system thinking choice, this new study added to the overall body of knowledge.

The research question of this new study asked what influence, if any, did instructions for discerning between intuitive and analytical decision tasks have on a person's choice of system thinking style for resolving both analytical and intuitive decision tasks. Would instructions for discerning between analytical and intuitive decision tasks influence a person's choice of thinking style? Would participants realize that a decision task was analytical and use their analytical thinking style to resolve it? Would participants realize that a decision task was intuitive and use their intuitive thinking style to resolve it? Those participants who were in the intervention group received four cognitive discerning instructions between analytical and intuitive decision tasks. Those intervention participants performed better in selection of the correct type of system thinking which applied to the different decision tasks. When the decision task type was analytical, intervention participants more frequently indicated that they selected the analytical thinking style for resolution than control participants. When the decision

task type was intuitive, intervention participants significantly selected more frequently the intuitive thinking style for resolution than control participants.

The null hypothesis was that the intervention group's mean for their choice of system one thinking style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically equal or less than the control group's mean for their choice of thinking style. As there was a significant difference, $t(108) = -2.36, p \leq .05$, between the control group's mean ($M = 65.42, SE = 2.45$) and intervention group mean ($M = 73.53, SE = 2.37$), the null hypothesis was rejected. If the intervention participants had not shown a significant difference then the null hypothesis would not have been rejected and the alternate hypothesis would have been rejected. This would indicate that the intervention was not supported which would lead to the conclusion that the intervention was ineffective.

The alternate hypothesis was that the intervention group's mean for their choice of system one thinking style for intuitive type decisions and system two thinking style for analytical type decisions would be statistically higher than the control group's mean for their choice of thinking style. The results supported the alternate hypothesis and provided support for the use of cognitive instructions for discernment as an intervention. On average, those participants who were provided with the intervention instructions for discerning between analytical and intuitive decision tasks scored higher on their correct choice of thinking style than those who did not have the instructions. This new study supported the use of cognitive instructions for discernment between analytical and intuitive decision tasks.

This study performed additional analysis to understand influences of other confounds such as participants' demographics. The analysis on gender, age, and education level did not show any significant difference. Separate t-tests were conducted for gender, age, and education examining only the control group, only the intervention group, and both groups. The results showed no significant difference between those of different gender, age, or education. Gender, age, and education did not appear to be a contributing factor to choice of system thinking style in these results. While intervention females showed a higher mean score than males it was not significant. While participants over age 39 performed higher than those below 40, the difference was not significant. While participants who completed high school only were higher in mean scores than those who completed college, the difference was not significant. These differences may become areas for researchers to investigate further in future studies which focus specifically in these demographic areas.

This study also separated out and analyzed the intuitive decision tasks from the analytical tasks which presented some interesting results. In analyzing only intuitive decision tasks, there was a significant difference between control and intervention groups' means. As prior research focused on instructions for analytical decision tasks and ignored intuitive type decisions (De Neys & Glumicic, 2008; Stuppel et al., 2013), this new study included both. Results supported the use of cognitive instructions for influencing choice of system thinking style in intuitive type decision tasks. In analyzing only analytical decision tasks, there was a difference between control and intervention groups' means but it was not significant. This needs further research as prior research supported the uses of cognitive instruction for analytical decision tasks. From these

results on intuitive decision tasks, these findings show a new discovery pertaining to the use of these instructions for influencing choice of intuitive thinking style for intuitive thinking decision tasks. While prior research assumed that intuitive decisions would not need instructions because they are intuitive, this study showed these instructions were a positive influence on choice of intuitive thinking style for intuitive decision tasks.

These results provided support fulfilling the purpose of addressing the problem of understanding how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for problem resolution. The intervention provided instructions for discerning between analytical and intuitive decision tasks. The survey included both analytical and intuitive decision tasks. The testing was on participants' choice of thinking style for problem resolution after resolving the decision tasks. The analysis evaluated the self-reported differences of thinking style between control and intervention groups' means. The results provided support for these four cognitive instructions as an intervention which included discernment for both intuitive and analytical decision tasks. Prior research did not investigate the influences of system one intuitive type instructions (Barton et al., 2009; De Neys & Glumicic, 2008; Kogler & Kühberger, 2007). This new study include both and provided a better understanding of how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for problem resolution.

Recommendations

This study revealed a new discovery on the effectiveness of these instructions for intuitive decision tasks. As the results supported the use of cognitive instructions for discerning between analytical and intuitive decision tasks, further research should

corroborate results before organizations provide these instructions for use by management. Management may find these instructions helpful in discerning between intuitive and analytical decision tasks. The expectation is that discernment should assist in managers' choice of system thinking style for problem resolution with the ultimate purpose of improving decision making. The expectation is that people will be better at decision making if they realize a decision task requires analytical thinking or that simple intuition is all that is needed. People are prone to error when their intuitive or analytical biases influence their decision making. They make errors when analytical biases result in over thinking or their intuitive biases result in impulsive unfounded decisions.

Organizational leaders would use this research to design training which provide these instructions to managers. Trainers would present the instructions in written, audio, and video form, reinforcing the need for discerning the difference between analytical and intuitive decision tasks. Trainers would explain the differences between analytical thinking and intuitive thinking and how these instructions would help managers know when to use which type of thinking for which type of decision task. Trainers would test managers as to their understanding of the instructions with practice question similar to the survey presented in this study. Managers would identify if they used analytical or intuitive thinking to resolve the decision tasks. Trainers would evaluate the responses to see if additional clarification was needed. Trainers would continue to work with managers presenting them with a summary card of the instructions which managers would take with them as they perform their regular work activities.

Though the results show a significant difference, there is still room for improvement in the effectiveness of these instructions. Further research should

experiment with audio instructions rather than written instructions. Results might have improved if participants could have heard rather than read the instructions. This study presented four instructions for discernment at the beginning of the survey after the informed consent form. These intervention instructions for identification of analytical decision tasks were used by Kogler and Kühberger (2007) and specified by Kahneman (2003). These four instructions involved looking for cues; considering vigilance and accuracy; understanding the amount of time needed for the response; and realizing if taking a second look would help. Participants may have not paid enough attention to the instructions after reading and acknowledging their willingness to participate during the informed consent process. Combining both audio and written may bring about better results.

Future research may include examples in the instructions or provide practice decision tasks before testing to increase the understanding of the discerning instructions. While results showed a significant difference for the intervention group, differences were higher for intuitive decision tasks than analytical decision tasks. Providing practice questions and examples may bring more clarity to the application of the instructions when testing. This would enable participants to first apply the instructions in practice without being scored. Participants would be able to repeat the practice session until they correctly chose the right style of thinking for the appropriate decision task. The practice session would include the instructions along with an equal number of intuitive and analytical decision tasks for resolution. Once the practice session was complete, participants would move on to the actual survey for scoring.

Future researchers may also want to keep the instructions visible during the testing for participants to reference. In this study, instructions were only visible at the beginning for the intervention participants and not visible at all for the control group. This method would still leave the choice of thinking style up to participants but would not require the participants to remember the instructions. Instructions could be placed at the top right of the screen in summary format reminding participants of the four instructions. This would not take away from the testing as the point is to understand the effectiveness of the instructions and not the participants' ability to remember and recall the instructions for application to the decision problems.

As results were different but not significant for analytical decision tasks, future researchers may want to use different and less complex analytical tasks. In this study, participants' answers to the decision tasks was not analyzed for decision accuracy because decision accuracy was not the focus. Participants may have mistakenly perceived the analytical decision tasks as intuitive because incorrect and correct answers were provided to participants. Because decision accuracy is not the focus, it may be better to use fill in the blank type questions rather than multiple choice. It may also have helped if each question did not have the option to mark it as cannot be determined. Using different analytical decision tasks which are less complex or not providing the answers may improve participants' choice of analytical system thinking style for analytical decision tasks. Each of these changes could be tested in future studies.

This researcher's focus was on managers but future research may want to include other job roles. This research works in an area where industrial and organizational psychology is extremely important with influencing senior leadership in a large

organization. Future researchers may wish to transform these cognitive instructions into easier reading levels or pictures for testing children. These decision tasks were designed for adults as were the instructions.

There may be value in understanding at what age level children are able to comprehend cognitive discernment instructions for decision tasks. Klossek et al. (2011) have experience with a single action and choice training group for children. They used an on-line test for the children which included cartoon videos as rewards. Ghetti and Angelini (2008) examined memory recall and recognition with children and adolescents in two experiments. Amsel et al. (2008) claimed a person's age and education increases his or her ability to control and utilize analytical thinking over system one. People are born with intuitive thinking ability and this ability continues to develop throughout their lives (Spezio, 2011). People develop their system one intuitive thinking through both conscious and unconscious means as they experience life through their impressions and feelings. People have to make a conscious and deliberate effort to develop and use system two analytical thinking style (Kahneman, 2011). The prefrontal cortex (PFC) is the location of the brain that controls analytical thinking (Spezio, 2011). From this part of the brain, people are able to reason, monitor their behavior, and take corrective actions in decision making. With this being the case, children will more easily resolve the intuitive decision tasks than the analytical ones as their analytical thinking must be developed as the PFC matures.

As results were significantly different for system thinking style choice of intuitive decision tasks, researchers may want to continue to use these decision tasks. Intuitive decision tasks used were designed to be questions which could be answered quickly

without much effort as used by De Neys and Glumicic (2008). Overall, this new research contributed to the use of cognitive instructions for discerning between intuitive and analytical decision tasks. These cognitive instructions had the most significant effect on intuitive choice for intuitive decision task than analytical thinking for analytical decisions.

Conclusions

This study investigated how instructions for discerning between analytical and intuitive tasks affected participants' choice of thinking style for problem resolution. Prior research had neglected instructions for discerning intuitive decision tasks and choice of intuitive thinking style (De Neys & Glumicic, 2008; Kogler & Kühberger, 2007). This study addressed this problem and answered the question in understanding how these discernment instructions influences participants choice of thinking style for both intuitive and analytical thinking styles. Findings supported that the intervention group's mean for their choice of system thinking style for decisions tasks were statistically higher than the control group's mean for their choice of thinking style. These results indicate that cognitive instructions as defined by Kahneman (2011) for discernment of decision tasks improved participants' choice of system thinking style. Future researchers may want to use other means to present cognitive instructions in addition to written instructions to improve results. As more research is conducted on these cognitive instructions, managers may find these instructions useful in discerning between analytical and intuitive decision tasks improving their choice of system thinking style. Prior research focused on analytical thinking while this new research included both analytical and intuitive system thinking in cognitive discernment instructions, decision tasks, and choice of system

thinking style which added to the overall body of knowledge for dual system theory (Thompson, Turner, et al., 2013).

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Appendixes

Appendix A: Literature Search Listing

Table 5
Search Criteria Listing with Count of Articles Used

Search Criteria	Articles Used
Analytical bias	3
Attentional bias	11
Cognitive control	43
Cognitive inhibition training	7
Cue reactivity	3
Debiasing	26
Deductive reasoning	51
Formulate an alternative	2
Influence of beliefs	7
Inhibit beliefs	4
Inhibition of belief	3
Inhibition training	10
Inhibitory control	30
Logical training	3
Mindware	13
Monitoring and control process	3
Presumption of rationality	2
Reasoning bias	29

Appendix B: Sample Survey

Page 1 Informed Consent

Informed Consent Form

Effects of Cognitive Instructions for Discerning Analytical and Intuitive Tasks

This purpose of this research is to learn how people think when given decision making instructions. This is an offer to you to help in research as part of the dissertation process at Northcentral University in Prescott, Arizona. You were chosen to be a volunteer through Qualtrics because you are an adult manager and you asked to take part in surveys.

Your part will be to answer 10 word problems by selecting a multiple choice answer. You may also choose not to answer. You may take as long as you wish. The survey takes about 15 minutes to complete. Participants will be randomly assigned to one of two groups, standard instructions or additional instructions.

Researchers involved are Donald High and Dr. Danielle Rowell. There are no known risks to you in being a part of this research. There are no personal questions. You may stop the survey at any time or simply not answer a question you do not feel like answering.

There are no known risks. These word problems have been used before in many other studies with no harm to others. Please try to solve each problem on your own. After each word problem, please choose how you went about solving the problem. You do not have to answer how. At any time, you may skip a question. You may quit the testing without any cost or penalty to you.

The benefits of this research may help others understand how directions help in resolving word problems. You may also benefit from this research as these directions are intended to help you and others make better decisions. This research does not require or use any of your personal information which might identify who you are.

All data collected will be kept confidential. No information which might identify who you are will be recorded for this research. You may review a summary once the research is complete. These results and instructions will be available to you on this survey site through the same link you used to access this survey. This information may help you in decision making.

Concerning your rights or complaints, you are free to contact the researchers of this research at D.High0771@email.ncu.edu or drowell@ncu.edu. If you have questions about your rights as a research participant, any complaints about your participation in the research study, or if any problems occurred in the study, please contact the researchers identified in the consent form. Or if you prefer to talk to someone outside the study team, you can contact Northcentral University's Institutional Review Board at irb@ncu.edu or 1-888-327-2877 ex 8014.

Being a part of this survey is voluntary. You do not have to answer any question. You

may quit at any time. Only Qualtrics will reward you with e-points for answering these questions.

If you wish to continue with the testing, press the accept button indicating you agree to the following and you are ready to begin answering questions;

- You have read all of these directions.
- You are volunteering for research.
- You are at least 18 years of age or older.

If you do not wish to continue with the testing, you press the cancel button without any cost or penalty to you.

Page 2 Discerning Instructions

Instructions for Discerning between Intuitive and Analytical Word Problems

The following are four instructions which will help you determine if a word problem can be easily answered intuitively or if you need more time to think it through:

- Is there **relevant information** in the word problem for the resolution?
 - When there are **relevant facts** that help, you will have to **think** about it.
 - When information is not relevant, you will rely on your intuition.
- How many **possible correct** answers are there?
 - When there is only **one correct answer**, you will have to **think** about it.
 - When any answer will do, you will rely on your intuition.
- How **much time** is needed for the response?
 - When you can **answer it quickly**, you will rely on your **intuition**.
 - When you have to take more time, you will have to think about it.
- Would a **second look** make a difference?
 - A second look **would not matter**, you will rely on your **intuition**.
 - A second look is needed, you will have to think about it.

Use these instructions as guidelines to resolve the word problems on the rest of the pages.

If you wish to continue with the testing, you may press the accept button which confirms the following and you will begin the survey process;

- You have read all the above instructions.
- You understand all the above instructions.

If you do not wish to continue with the testing, you may press the cancel button without any consequences.

- Accept
- Cancel

Page 3 Erin

1. Erin is 13 years old. Erin's favorite subject is art. Erin's favorite things to do are shopping and having sleepovers with friends to gossip about other kids at school. What is most likely? Click on answer below:
 - a. Erin is a girl
 - b. Erin is a boy
 - c. Cannot be determined
2. To answer this question, which of the following did you do? Click on answer below:
 - a. Used intuition
 - b. Had to think things through carefully
 - Skip the question
 - Exit survey

Page 4 Jay

3. Jay is a 29-year-old male. He has served a short time in prison. He has been living on his own for 2 years now. He has an older car and listens to punk music. What is most likely? Click on answer below:
 - a. Jay has a tattoo
 - b. Jay has no tattoo
 - c. Cannot be determined
4. To answer this question, which of the following did you do? Click on answer below:
 - a. Used intuition
 - b. Had to think things through carefully
 - Skip the question
 - Exit survey

Page 5 Lilly

5. Lilly is 37 years old. She is married and has 3 kids. Her husband is a veterinarian. She is committed to her family and always watches the daily cartoon shows with her kids. What is most likely? Click on answer below:

- a. Lilly is a kindergarten teacher
 - b. Lilly is an executive manager
 - c. Cannot be determined
6. To answer this question, which of the following did you do? Click on answer below:
- a. Used intuition
 - b. Had to think things through carefully
- Skip the question
 - Exit survey

Page 6 Tara

7. Tara is 15. She loves to go shopping at the mall and to talk with her friends about their crushes at school. What is most likely? Click on answer below:
- a. Tara is a Bruce Springsteen fan
 - b. Tara is a Britney Spears fan
 - c. Cannot be determined
8. To answer this question, which of the following did you do? Click on answer below:
- a. Used intuition
 - b. Had to think things through carefully
- Skip the question
 - Exit survey

Page 7 Martine

9. Martine is 26 years old. She is bilingual and reads a lot in her spare time. She is a very fashionable dresser and a great cook. What is most likely? Click on answer below:
- a. Martine is American
 - b. Martine is French
 - c. Cannot be determined
10. To answer this question, which of the following did you do? Click on answer below:
- a. Used intuition
 - b. Had to think things through carefully
- Skip the question

- Exit survey

Page 8 Bat and Ball

11. A bat and a ball cost \$1.10 in total. The bat costs \$1.00 more than the ball. How much does the ball cost? Click on answer below:
- 10 cents
 - 5 cents
 - Cannot be determined
12. To answer this question, which of the following did you do? Click on answer below:
- Used intuition
 - Had to think things through carefully
- Skip the question
 - Exit survey

Page 9 Widgets

13. If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets? Click on answer below:
- 5 minutes
 - 100 minutes
 - Cannot be determined
14. To answer this question, which of the following did you do? Click on answer below:
- Used intuition
 - Had to think things through carefully
- Skip the question
 - Exit survey

Page 10 Lily Pads

15. In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover half of the lake? Click on answer below:
- 24 days
 - 47 days
 - Cannot be determined

16. To answer this question, which of the following did you do? Click on answer below:

- a. Used intuition
- b. Had to think things through carefully
- Skip the question
- Exit survey

Page 11 Jack, Anne, and George

17. Jack is looking at Anne but Anne is looking at George. Jack is married but George is not. Is a married person looking at an unmarried person? Click on answer below:

- a. Yes
- b. No
- c. Cannot be determined

18. To answer this question, which of the following did you do? Click on answer below:

- a. Used intuition
- b. Had to think things through carefully
- Skip the question
- Exit survey

Page 12 Bicyclist

19. A bicyclist completes a ten-mile race track with a slow average speed of 10 miles/hour. How fast should the bicyclist cycle on the second round to reach a more impressive average speed of 20 miles/hour on both rounds? Click on answer below:

- a. 30 miles/hour speed
- b. 40 miles/hour speed
- c. Cannot be determined

20. To answer this question, which of the following did you do? Click on answer below:

- a. Used intuition
- b. Had to think things through carefully
- Skip the question
- Exit survey

Page 13 Demographics

21. What is your age:

a. _____

22. What is your gender:

a. Male

b. Female

c. Skip the question

23. What is your education level completed:

a. High school

b. College

c. Graduate

d. Skip the question

e. Exit survey

Page 14 Thank You

Thank you for your time and for your consideration of this research on intuitive and analytical thinking. Once the study is completed you may return to this page to see the results and any instructions used in this research for your benefit.